Contents

1 Differentiable structures on topological manifolds

This section aims to set the playing field for the thesis. Due to the theory being of intrest for mathematicians and physisists alike there are many (equivalent) definitions used to introduce the objects. The main goal of this section is to avoid confusion, by giving the definitions instead relying on the reader to guess what a certain symbol stands for.

Definition 1.0.1 (Topological Manifold)**.** A second countable Hausdorffspace M is called **topological manifold** of dimension $m \in \mathbb{N}$, if it is locally homeomorphic to $\mathbb{R}^m$. To be precise, if for all $p \in M$ there exists an open neighborhood $U \subseteq M$ of p, an open set $V \subseteq \mathbb{R}^m$ and a map $\varphi : W \to V$ that is a homeomorphism. We call the map $\varphi : U \to V$ a **chart around p on M** and φ^{-1} a **local coordinate system around p on M**.

Definition 1.0.2 (Differentiable Manifold)**.** Let M be a topological manifold of dimension M.

1. A **differentiable atlas of class** $r \in \mathbb{N} \cup \{\infty\}$ is a family of charts $\mathfrak{A} = (\varphi_i : U_i \to V_i)_{i \in I}$ such that
   a) $\bigcup_{i \in I} U_i = M$, meaning that (U_i) is an open covering of M.
   b) For every pair $(i, j) \in I^2$ the **transition function**:
   $$\begin{aligned} \varphi_{ij} : \varphi_j(U_i \cap U_j) &\to \varphi_i(U_i \cap U_j) \\ x &\mapsto \left(\varphi_i \circ \varphi_j^{-1}\right)(x) \end{aligned}$$
   is differentiable of class r.

   We call such an atlas a C^r-atlas.
2. Two C^r-atlases $\mathfrak{A}$ and $\mathfrak{B}$ are called **equivalent** if the family $\mathfrak{A} + \mathfrak{B} = (\varphi_i, \varphi_j)_{ij}$ is a C^r-atlas.

A **differentiable structure of class** r on M is an equivalence class c of C^r-atlases. For $r = \infty$ we call the pair (M, c) a smooth manifold.

Corollary 1.0.3. Every transition functions φ_{ij} $i, j \in I^2$ of a differentiable atlas $\mathfrak{A} = (\varphi_i)_{i \in I}$ is not just a homeomorphism but also a diffeomorphism due to $\varphi_{ji} = \varphi_{ij}^{-1}$

Definition 1.0.4. Let (M, c) be a differentiable manifold of class r and $U \subseteq M$ open. We call a continuos function

$$f : U \to \mathbb{R}$$

differentiable of class r, if for any one (and hence for all) $(\varphi_i)_{i \in I} = \mathfrak{A} \in c$ the compositions $f \circ \varphi_i^{-1}$ are differentiable of class r. For $r = \infty$ we define:

$$\mathcal{E}(U) = \{ f \in U \to \mathbb{R} \text{ continouos} \mid f \text{ is differentiable of class } \infty \}.$$

Corollary 1.0.5. Let (M,c) be a smooth manifold of dimension m and $U \subseteq M$ be a open subset. With pointwise defined operations, the set $(\mathcal{E}(U),+,\cdot,\circ)$ becomes an $\mathbb{R}$–algebra. Furthermore, $\mathcal{E}$ becomes a sheaf of $\mathbb{R}$-algebras.

Proof. There is not really a need for a proof. However, it might help to work through the definition of a sheaf as a reminder. First, $\mathcal{E}$ is a presheaf, where the restriction in the domain of a function gives the needed restriction homomorphism:

$$\begin{aligned}\mathrm{res}^U_V : \mathcal{E}(U) &\to \mathcal{E}(V)\\ f &\mapsto f|_V \,.\end{aligned}$$

The required properties of a presheaf are trivial. Furthermore, this gives a sheaf as the requirement of locality is trivial for functions and the property of gluing is also trivial for functions, since differentiability is a local property. □

Definition 1.0.6. If $p \in M$ is fix, $f \in \mathcal{E}(U)$ and $g \in \mathcal{E}(U')$ such that $p \in U \cap U'$ we say that f and g have the same **germ in** p, if there is another open neighborhood $W \subseteq U \cap U'$ of p such that $f|_w = g|_W$. This defines an equivalence relation $\sim_p$. An equivalence class s of local functions around p is called a **germ in** p. We write $s = f_p$, if $s = [f]$ with $f \in \mathcal{E}(U)$. We write

$$\mathcal{E}_p(M) = \left(\sum_{U \text{open}, p \in U} \mathcal{E}(U)\right) \Big/ \sim_p \,.$$

For the set of germs and call it the **stalk in** p. Here $\sum$ denotes the co-product (also called sum) in **Top** and hence the disjoint union.

Corollary 1.0.7. For a smooth manifold (M,c) the set $\mathcal{E}_p(M)$ inherits an $\mathbb{R}$–algebra structure from the $\mathcal{E}(U)$. Furthermore, it carries a natural (evaluation-)homomorphism:

$$\begin{aligned}\mathrm{eval}_p : \mathcal{E}_p(M) &\to \mathbb{R}\\ f_p &\mapsto f(p) =: f_p(p)\end{aligned}$$

The stalks are also local rings with maximal ideal $\mathfrak{m}_p = \ker(\mathrm{eval}_p)$. Hence, the pair $(M,\mathcal{E})$ gives us a locally ringed space.

Proof. Here, we only need to prove the statement about the locality of the stalks. This follows from $f_p \in \mathcal{E}_p(M)$ being invertible if and only if $f(p) \neq 0$ which is the same as $f_p \notin \ker(\mathrm{eval}_p)$. □

Definition 1.0.8. Let (M,c) be a smooth manifold of dimension m and $p \in M$. We call an $\mathbb{R}$-linear map $\delta : \mathcal{E}_p(M) \to \mathbb{R}$ a **derivation**, if it satisfies the Leibnitz-rule:

$$\delta(f_p \cdot g_p) = \delta(f_p) g_p(p) + f_p(p)\delta(g_p) \quad \text{for all } f,g \in \mathcal{E}_p(M)\,.$$

We call $\mathrm{Der}_{\mathbb{R}}(\mathcal{E}_p(M),\mathbb{R})$ the set of derivations and give it a $\mathbb{R}$ vector space structure by pointwise operations. We define the **tangent space of** M **at** p to be the vector space

$$TM_p := \mathrm{Der}_{\mathbb{R}}(\mathcal{E}_p(M),\mathbb{R})\,.$$

Corollary 1.0.9. Let (M,c) be a smooth manifold and $\varphi : U \to V$ be a chart around p with $x_0 = \varphi(p)$ $(\varphi \in \mathfrak{A} \in c)$. Then

$$\xi = \left.\frac{\partial}{\partial x_j}\right|_p : \mathcal{E}_p(M) \to \mathbb{R}$$

$$f_p \mapsto \xi(f_p) = \left.\frac{\partial}{\partial x_j}\right|_{x_0} (f \circ \varphi^{-1})$$

is well-defined and a tangent vector. In fact, the family

$$\left(\left.\frac{\partial}{\partial x_1}\right|_p, \dots, \left.\frac{\partial}{\partial x_m}\right|_p \right)$$

defines a basis of TM_p. Hence, the dimension of TM_p is m.

Definition 1.0.10. For a smooth manifold (M,c) the sum $\sum_p TM_p$ comes with a natural projection

$$\pi : TM \to M$$

$$\xi \mapsto p \text{ where } \xi \in TM_p$$

Furthermore, the **local vector fields** with respect to a chart $\varphi : U \to V$

$$\frac{\partial}{\partial x_j} : U \to \pi^{-1}(U)$$

$$p \mapsto \left.\frac{\partial}{\partial x_j}\right|_p$$

induce a local trivialization:

$$\pi^{-1}(U) \cong U \times \mathbb{R}^m .$$

We can induce a topology on TM such that all those trivializations are continuos. This then gives an atlas for TM such that we have a $2m$-dimensional manifold. To be precise, the atlas is given by the maps $\pi^{-1}(U_i) \to R^m \times R^m; x \mapsto (\pi(x), q_{\varphi \circ \pi(x)}(x))$ where q_p denotes the coordinate map corresponding to the basis $(\left.\frac{\partial}{\partial x_1}\right|_p, \dots, \left.\frac{\partial}{\partial x_m}\right|_p)$ that depends on the chart φ_i. In fact, this yields a smooth manifold and a (smooth) vector bundle of dimension m. We call TM the **tangent bundle**.

Definition 1.0.11 (The Derivative)**.** Let (M,c) and (M',c') be smooth manifolds (from now on we suppress the differentiable structure in our notation). We call a continuous function $f : M \to M'$ **smooth**, if for all $\varphi \in \mathfrak{A} \in c$ and $\varphi' \in \mathfrak{A}' \in c'$ the maps

$$\varphi' \circ f \circ \varphi^{-1} : V \to V'$$

are smooth. A given smooth function induces a smooth function between the tangent bundles as follows:

$$Df : TM \to TM' \;, Df_p(\xi)(g_p) = \xi((g \circ f)_p)$$

Here, $\xi \in T_pM$, $g_p \in E_p(M')$.

Corollary 1.0.12. Let $f : M \to \mathbb{R}$ be smooth and $\varphi : U \to V$ be a chart. Then we can interpret $\mathrm{d}f$ as a one-form. To be precise assume q to be the coordinate function $T\mathbb{R} \to \mathbb{R}$ from the basis induced by the identity as a chart. $v \in \Gamma TM$ we have

$$q \circ \mathrm{d}f(v) = v \cdot f$$

Here on the right side, f denotes a map $p \mapsto f_p$ such that $v(f)(p) := v_p(f_p)$ is well defined. We keep this notations so vector fields can take functions as an input.

Proof. We proof this by showing it for the section $\frac{\partial}{\partial x_i}$ and thereby for any, since those sections form a basis of the space of sections as a $C^\infty(M, \mathbb{R})$ vector space. Now let $g_p \in \mathcal{E}_p(\mathbb{R})$ and $\varphi(p) = x_0$. Then

$$\mathrm{d}f(\frac{\partial}{\partial x_i}\bigg|_p)(g_p) = \frac{\partial}{\partial x_i}\bigg|_p (g \circ f)_p = \frac{\partial}{\partial x_i}\bigg|_{x_0} (g \circ f \circ \varphi^{-1}) = \frac{\partial}{\partial x}\bigg|_p (g_p) \cdot \frac{\partial}{\partial x_i}\bigg|_p (f_p)$$

Hence, $\mathrm{d}f(v) = v(f)\frac{\partial}{\partial x}$ letting us conclude the statement. □

Definition 1.0.13 (A Metric)**.** Let M be a smooth manifold. A section $g \in \Gamma(TM^* \otimes TM^*)$ into the tensor product of the dual tangend bundle with itself is a **Riemannian metric** if the following are satisfied for all $p \in M$:

1. g is **non degenerate**, meaning for a $v_p \in T_pM$ $g_p(v_p, w_p) = 0$ $\forall w_p \in T_pM$ then $v_p = 0$.
2. g is **symmetric**, meaning $g_p(v_p, w_p) = g_p(w_p, v_p)$ $\forall v_p, w_p \in T_pM$.
3. g is **positive definite**, meaning that $g_p(v_p, v_p) \geq 0 \forall v_p$ and vanishes only for $v_p = 0$.

We cal a tuple (M, g) a **Riemannian manifold**.

Definition 1.0.14 (The Gradient)**.** Assume that (M, g) is a smooth manifold together with a Riemannian metric. Let $f : M \to \mathbb{R}$ be a smooth function. We define its **gradient** to be a vector field $\mathrm{grad}(f) \in \Gamma(TM)$ such that for any vecor field V:

$$q \circ \mathrm{d}f(V) = g(\mathrm{grad}(f), V)\,.$$

Here q denotes the canonical coordinate function $T\mathbb{R} \to \mathbb{R}$.

Definition 1.0.15. Let (M, g) be a Tiemannian manifold and $\varphi : U \to V$ be a chart. We define the smooth functions $g_{ij} : U \to \mathbb{R}$ to be :

$$g_{ij} = g(\frac{\partial}{\partial x_i}, \frac{\partial}{\partial x_j})\,.$$

Then if two vector fields v, w over U are given with $v = \sum v^i \frac{\partial}{\partial x_i}$ and $w = \sum_i w^i \frac{\partial}{\partial x_i}$ we can calculate the metric locally:

$$g(v, w) = \sum_{ij} g_{ij} v^i w^j \, : U \to \mathbb{R}$$

Furthermore, we can invert the matrices $(g_{ij}(p))_{ij}$ for all p (which is a smooth procedure which can be seen by the construction via Cramers rule) to get smooth functions

$$\begin{aligned} g^{ij} : U &\to \mathbb{R} \\ p &\mapsto g^{ij}(p) = \left((g_{kl}(p))^{-1}_{kl}\right)_{ij} \end{aligned}$$

Lemma 1.0.16. *Let $\varphi : U \to V$ be a chart. Then the gradient has the local form:*

$$\mathrm{grad}(f) = \sum_{i,j} g^{ij}\left(\frac{\partial}{\partial x_i}(f)\right)\frac{\partial}{\partial x_j} =: \sum_{i,j} g^{ij}\frac{\partial f}{\partial x_i}\frac{\partial}{\partial x_j}.$$

Proof. Assume that $v, w \in \Gamma(TM)$ such that $v = \sum_i v^i \frac{\partial}{\partial x_i}$ and $w = \sum_i w^i \frac{\partial}{\partial x_i}$. Then $g(v,w) = \sum_{i,j} g_{ij} v^i w^i$. Suppose that $\mathrm{grad}(f) = \sum_j G^j \frac{\partial}{\partial x_j}$ and q is the coordinate map $T\mathbb{R} \to \mathbb{R}$ in each tangend space. Then by definition of the gradient we have:

$$\frac{\partial}{\partial x_j}(f) = q \circ \mathrm{d}f(\frac{\partial}{\partial x_j}) = q \circ \frac{\partial f}{\partial x_j} = g(\mathrm{grad}(f), \frac{\partial}{\partial x_j}) = \sum_{ij} g_{ij} G^i$$

Hence,

$$\begin{aligned} (G^1, \cdots G^m)(g_{ij}) &= \left(\frac{\partial}{\partial x_1}(f), \cdots, \frac{\partial}{\partial x_m}(f)\right) \\ \Leftrightarrow (G^1, \cdots G^m) &= \left(\frac{\partial}{\partial x_1}(f), \cdots, \frac{\partial}{\partial x_m}(f)\right)(g^{ij}). \end{aligned}$$

But then we can conclude that $G^j = \sum_i g^{ij}\frac{\partial}{\partial x_i}(f)$.

□

Definition 1.0.17 (A Dynamical System)**.** Let M be a smooth manifold. A **dynamical system on** M is given by a smooth map $\varphi : \Omega \to M$ such that:

1. $\{0\} \times M \subseteq \Omega \subseteq \mathbb{R} \times M$ open, and for any $p \in M$ the set $I(p) := \pi_1(\Omega \cup \mathbb{R} \times \{p\})$ is connected. (π_1 is the projection onto the first factor)

2. For all $p \in M$ and $t \in I(p)$ we have that $s \in I(\varphi(t,p))$ if and only if $s + t \in I(p)$. Then we have:

$$\varphi(s, (\varphi(t,p))) = \varphi(s+t, p)$$

3. For all p we require $\varphi(0,p) = p$.

We will write $\varphi_t(p) := \varphi(t,p)$ to keep the notation bearable. If for all p $I(p) = \mathbb{R}$ we call the system **global**.

Corollary 1.0.18. For a global dynamical system φ on a smooth manifold M the map

$$\rho : \mathbb{R} \to \mathrm{Diff},\ t \mapsto \varphi_t$$

is a homomorphism. To see the well definition notice how for all $t \in \mathbb{R}$ φ_{-t} is an inverse of φ_t.

Definition 1.0.19 (Vector Field of a Dynamical System)**.** Let φ be a dynamical system on a smooth manifold M. Then we call the mapping

$$p \mapsto X(p) := \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} \varphi_t(p) \in T_pM$$

the **associated vecor field**. This is in fact a smooth vector field.

Definition 1.0.20 (Dynamical System of a Vector Field)**.** Let $X \in \Gamma(M)$ be a global vector field on the smooth manifold M. Then we call a dynamical system φ on M the **associated dynamical system** to X, if for all $t \in \mathbb{R}$ and $x \in M$

$$\mathrm{d}_t \varphi_t(x)|_{t=t_0} = X(\varphi_{t_0}(x))\,.$$

d_t denotes the differential in t, meaning for fixed x we differentiate

$$\varphi(x) : \mathbb{R} \to M, \quad t \mapsto \varphi_t(x)$$

and evaluate that in the constant trivial vector field $E : \mathbb{R} \to T\mathbb{R}$ with $E(t) = 1_t$.

Whilst differentiating leads to a smooth vector field, we can also go the other direction by locally integrating a vector field to get a flow. To be precise: Those concepts are inverse to one another or in other words:

Theorem 1.0.21. *Let M be a smooth manifold. Then there is a one to one correspondence between global maximal flows and vector fields.*

Definition 1.0.22. Let V be a linear manifold (i.e. a finite dimensional vector space). Then for any $v \in V$ we have the canonical maps

$$\begin{aligned} Q_v : V &\to T_vV \\ x &\mapsto \left(f_v \mapsto \left.\frac{\partial}{\partial t}\right|_{t=0} f(v+tx)\right) \end{aligned}$$

Let $L : V \to V$ be a linear map viewed as a smooth map between manifolds. Then we have the identity:

$$\mathrm{d}L|_v \circ Q_v = Q_{L(v)} \circ L\,.$$

This is proven by a short calculation:

$$\Big(\,\mathrm{d}L|_v\,\big(Q_v(x)\big)\Big)(f_{L(v)}) = Q_v(x)\,(f\circ L) = \frac{\partial}{\partial t}\Big|_{t=0} f\circ L(v+tx) = \underbrace{\frac{\partial}{\partial t}\Big|_{t=0} f\big(L(v)+tL(x)\big)}_{=Q_{L(v)}\circ L(x)}$$

Now let $v, v' \in V$ be two points in V. Then we have the translation

$$tr_{v,v'} : V \to V;\ x \mapsto (x + (v' - v))$$

With this we can now find a connection between Q_v and $Q_{v'}$ as follows:

$$\big(\mathrm{d}tr_{v,v'}|_v\,(Q_v(x))\big)\,(f_{v'}) = \frac{\partial}{\partial t}\Big|_{t=0} (f\circ tr_{v,v'}(v+tx)) = \frac{\partial}{\partial t}\Big|_{t=0} \big(f(v'+tx)\big) = Q_{v'}(x)(f_{v'})\,.$$

Hence:

$$Q_{v'} = \mathrm{d}tr_{v,v'}\circ Q_v\,.$$

Remark 1.0.23. Let $X \in \Gamma(TM)$ be a vector field on the smooth manifold M, and φ be the associated dynamical system. For any fixed $t_0 \in \mathbb{R}$ and any $x_0 \in M$ we have the identity

$$\mathrm{d}\varphi_{t_0}|_{x_0}\circ (X(x_0)) = X(\varphi_{t_0}(x_0))\,.$$

Proof. This follows directly from the second property of flows as follows: We first make use of the identity $\varphi_{t_0}(x_0) = \varphi_{t_0}\circ\varphi_0(x_0)$ and $X(x_0) = \mathrm{d}_t\varphi_t(x_0)|_{t=0}$ to calculate using the chain rule:

$$\mathrm{d}_t\varphi_{t_0+t}(x_0)|_{t=0} = \mathrm{d}_t\varphi_{t_0}\circ\varphi_t(x_0)|_{t=0} = \mathrm{d}_x\varphi_{t_0}|_{x=\varphi_0(x_0)}\circ \mathrm{d}\varphi_t(x_0)|_{t=0} = \mathrm{d}_x\varphi_{t_0}|_{x=x_0}\circ X(x_0)\,,$$

and alternatively

$$\mathrm{d}_t\varphi_{t_0+t}(x_0)|_{t=0} = \mathrm{d}_t\varphi_t(\varphi_{t_0}(x_0))|_{t=0} = X(\varphi_{t_0}(x_0))\,.$$

□

2 Morse Theory

For everything concerning Morse theory our main reference will be the wonderful book [BH04]. In Morse Theory, we will always deal with a smooth function $f : M \to \mathbb{R}$ from a smooth Riemannian manifold (M, g) that is also compact. We will denote its dimension by $m = \dim(M)$.

Definition 2.0.1 (Critical Points)**.** We call a point $p \in M$ **critical**, if $\mathrm{d}f|_p$ vanishes meaning that

$$\mathrm{d}f|_p : T_pM \to T_{f(p)}\mathbb{R}$$

is trivial. By $\mathrm{Crit}(f)$ we denote the **set of critical points**.

Definition 2.0.2 (The Hessian)**.** Let $f : M \to \mathbb{R}$ be a smooth function. The **Hessian** of f at $p \in \mathrm{Crit}(f)$ is a symmetric bilinear function

$$\mathrm{Hess}_p(f) : T_pM \times T_pM \to \mathbb{R}$$

that is defined as follows. Let $\xi_p, \zeta_p \in T_pM$ be two tangent vectors. Now choose two vector fields Ξ and Z in a neighbourhood of p such that $\Xi(p) = \xi_p$ and $Z(p) = \zeta_p$. With this define the Hessian to be:

$$\mathrm{Hess}_p(f)(\xi_p, \zeta_p) = (\Xi \cdot (Z \cdot f))\,(p)$$

Compare 1.0.12 for the notation.

Definition 2.0.3. A smooth function $f : M \to \mathbb{R}$ is called **Morse**, if for any critical point.

The Hessian is a symmetric bilinear and non-degenerate Map. For such a map there exists the following theorem:

Theorem 2.0.4 (Sylvesters Law of Inertia)**.** *Let $A \in \mathrm{Mat}(n, \mathbb{R})$ be symmetric and let $T, T' \in \mathrm{GL}(n, \mathbb{R})$ and $k, k', l, l' \in \mathbb{N}$ such that*

$$T^t \circ A \circ T = \begin{pmatrix} \mathbb{1}_k & 0 & 0 \\ 0 & -\mathbb{1}_l & 0 \\ 0 & 0 & 0_{n-k-l} \end{pmatrix} \textit{ and } T'^t \circ A \circ T' = \begin{pmatrix} \mathbb{1}_{k'} & 0 & 0 \\ 0 & -\mathbb{1}_{l'} & 0 \\ 0 & 0 & 0_{n-k'-l'} \end{pmatrix}$$

then $k = k'$, $l = l'$ and $\mathrm{rank}(A) = k + l$.

Proof. Since T, T' are invertible we have that

$$k + l = \mathrm{rank}(T^t \circ A \circ T) = \mathrm{rank}(A) = \mathrm{rank}(T'^t \circ A \circ T') = k' + l'\,. \tag{1}$$

Hence it suffices to show that $k = k'$. Which we will do by proofing the claim:

$$k = \max\left\{\dim(U) | U \subseteq \mathbb{R}^n \text{subspace such that } x^t A x > 0 \ \forall x \in U \setminus \{0\}\right\}$$

So we start by showing „≤“: Denote the first k colons of T with $x_1, ..., x_k$. They form a basis of $\mathbb{R}^n$ and with $0 \neq x = \sum_{i=1}^k \lambda i x_i$ we have that by bilinearity

$$x^t A x = \sum_{i=1}^k \lambda_i x_i^t A x = \sum_{i,j=1}^k \lambda_i \lambda_j x_i^t A x_j = \sum_{i=1}^k (\lambda_i)^2 \geq 0\,. \tag{2}$$

This concludes the first inequality.

Now let U be any $k-$dimensional subspace such that for all non zero $x \in U$ $x^t A x > 0$. By a calculation analog to the one above we have that for any $x \in W := \operatorname{span}(x_{k+1}, \ldots x_n)$ the number $x^t A x$ is less or equal to zero. Hence, $W \cap U = \{0\}$ and with this we conclude:

$$\dim(U) = \dim(U + W) - \dim(W) + \dim(U \cap W) \leq (k + (n-k)) - (n-k) + 0 = k\,. \tag{3}$$

This is the last inequality proving the statement. □

Corollary 2.0.5. Let $\varphi : U \to V$ be a chart around a critical point $p \in M$. Then in said chart the matrix corrsponding to the Hessian has the form:

$$M_p(f) = \left(\frac{\partial^2 (f \circ \varphi^{-1})}{\partial x_i \partial x_j}\right)_{i,j}$$

Proof. This is just an easy calculation, since $\operatorname{Hess}_p(f)\left(\left.\frac{\partial}{\partial x_i}\right|_p, \left.\frac{\partial}{\partial x_j}\right|_p\right)$ is given by

$$\left(\left.\frac{\partial}{\partial x_i}\right|_p \left(\frac{\partial}{\partial x_j}(f)\right)_p\right) = \left(\frac{\partial^2 (f \circ \varphi^{-1})}{\partial x_i \partial x_j}\right)$$

□

The Hessian matrix $M_p(f)^i$ in two different charts $\varphi_i : U_i \to V_i$ for $i = 1, 2$ transforms via a congruence transformation by the Jacobian of the transition function. If we have two different charts. Then the Matrix $M_p(f)$ of the Hessian transformes from one base to the other via conjugation of the differential of the transition function $\varphi_{12} := \varphi_1 \circ \varphi_2^{-1}$:

$$M_p^1(f) = \mathrm{d}\varphi_{12}^t \cdot M_p^2(f) \cdot \mathrm{d}\varphi_{12}$$

Hence we can define the following:

Definition 2.0.6. By Sylvesters Law of interta, the number l that denotes the dimension of the maximal subspace on which a matrix is negative definite, is a invariant under conjugation. We call that number the index of said matrix. Now in the setting of Morse theory we assign to each kritical point p the index of $M_p(f)$ and call that the **(Morse-)index** of p, denoted with λ_p. If we cannot suppres the function without loss of clearity we will write λ_p^f. We denote the set of all critical points of index k by $\operatorname{Crit}_k(f)$.

Theorem 2.0.7 (Morse Lemma)**.** *Let p be a critical point of index k of the Morse funktion $f : M \to \mathbb{R}$. There exists a centered chart $\varphi : U \to \mathbb{R}^m$ arround p such that*

$$(f \circ \varphi^{-1})(x_1, \dots, x_m) = f(p) + \sum_{i=1}^{k} -x_i^2 + \sum_{i=k+1}^{m} x_i^2$$

Proof. Compar for example Lemma 3.11 in [BH04]. □

hier noch mehr Quellen!

Definition 2.0.8. Let $\varphi_t : M \to M$ be the local one-parameter group of diffeomorphisms generated by $-\mathrm{grad}(f)$, i.e. if $e : \mathbb{R} \to T\mathbb{R}$ denotes the canonical unit vector field that comes from the identity as the global chart on $\mathbb{R}$, we have that

$$\frac{\mathrm{d}}{\mathrm{d}t}\varphi_t(x) \circ e = -\mathrm{grad}(f)(\varphi_t(x)), \quad \text{and}$$
$$\varphi_0(x) = x\,.$$

We call that group the **negative gradient flow** of f with respect to the metric g. The integral curve $\gamma_x : (a, b) \to M$ given by

$$\gamma_x(t) = \varphi_t(x)$$

is called a gradient **flow line**.

Next we will deepen our **local** understanding of said vector fiel. Notice how the Morse Lemma is a statement about smooth manifolds-no metric needed nor mentioned. In fact there are many Morse charts and we can finde one, where the riemannian metric in p is diagonal with non-zero. To proof such a statement we first need a somewhat obvious lemma:

Lemma 2.0.9. *Assume that $\psi : U \to V$ is a chart and $L : R^m \to R^m$ a linear function. Denote the local coordinate maps of the Tangend bundle in a point p that comes from a basis in said point induced from a chart χ by q_χ. Then the diagramm*

$$\begin{tikzcd}
T_pM \arrow[r, "q_\psi|_p"] \arrow[dr, "q_{L\circ\psi}|_p"'] & \mathbb{R}^m \arrow[d, "L"] \\
& \mathbb{R}^m
\end{tikzcd}$$

commutes for all $p \in U$.

Proof. We show this by showing the inverse: For any $v \in \mathbb{R}^m$ we have the two $(q_\psi\big|_p)^{-1} \circ L^{-1}(v)$ and $(q_{\psi\circ L}\big|p)^{-1}$ and claim that they are the same. Assume that $L^{-1} = (\lambda_{ij})_{ij}$, $\psi(p) = x_0$, $L(x_0) = \tilde{x_0}$, $(v_1, \dots v_m) = v \in \mathbb{R}^m$ and denote the basis of T_pM coming from the chart ψ by $\left(\frac{\partial}{\partial x_i}\big|_p\right)_i$ and the one coming from $L^{-1} \circ \psi$ by $\left(\frac{\partial}{\partial \tilde{x}_i}\big|_p\right)_i$. We now calculate for $f_p \in \mathcal{E}_p M$:

$$\begin{aligned}
\left[(q_{\psi \circ L}\big|_p)^{-1}(v)\right](f_p) &= \sum_i v_i \frac{\partial}{\partial \tilde{x}_i}\Big|_p (f_p) \\
&= \sum_i v_i \frac{\partial}{\partial x_i}\Big|_{\tilde{x}_0} (f \circ \psi^{-1} \circ L^{-1}) \\
&= \sum_i v_i \sum_j \frac{\partial f \circ \psi^{-1}}{\partial x_j}\Big|_{x_0} \cdot \frac{\partial L_j^{-1}}{\partial x_i}\Big|_{\tilde{x_0}} \\
&= \sum_{i,j} v_i \frac{\partial}{\partial x_j}\Big|_p (f_p) \cdot \lambda_{ji} \\
&= \sum_{i,j} v_i q_\psi\big|_p^{-1}(e_j)(f_p) \cdot \lambda_{ji} \\
&= \left[q_\psi\big|_p^{-1}\big(\sum_{i,j} v_i \lambda_{ji} e^j\big)\right](f_p) \\
&= \left[q_\psi\big|_p^{-1}(L^{-1}(v))\right](f_p)
\end{aligned}$$

This concludes the proof. □

Now we can proof that a Morse chart exists where the gradient is of particular easy form:

Theorem 2.0.10 (Simultaneous Diagonalisation of Quadratic Forms)**.** *Let p be a critical point of a Morse function f on a manifold M and let $g(p)$ be the Riemannian metric on T_pM. Then there exists a Morse chart ψ arround p such that the representation of $g(p)$ in the basis induced by the Morse chart is given by a a diagonal matrix $diag(\mu_1, \dots \mu_m)$ with $\mu_i > 0$ for all i .*

Proof. The quadratic form of $f \circ \psi^{-1} - f(p)$ in the coordinates of any Morse chart ψ is $q_H(v) = v^T H v$. The quadratic form induced by the Riemannian metric $g(p)$ is $q_G(v) = v^T G v$, where $v \in \mathbb{R}^m$ are the coordinate vectors with respect to the Morse chart. To be precice this means for $v, w \in \mathbb{R}^m$:

$$g \circ (q_\psi\big|_p \oplus q_\psi\big|_p)(v, w) = v^T G w \text{ and } f \circ \psi^{-1}(v) = f(p) + v^T H v\,.$$

We now aim to manipulate ψ such that G becomes diagonal: Since $g(p)$ is positive definite, G is also positive definite, and H is symmetric.

Consider the generalized eigenvalue problem $Hv = \lambda G v$. Since H and G are real symmetric matrices and G is positive definite, this problem has m real eigenvalues $\lambda_1, \dots, \lambda_m$ and corresponding eigenvectors $v_1, \dots, v_m$, which can be chosen to be orthogonal with respect to the bilinear form defined by G, such that $v_i^T G v_j = \delta_{ij}$, since if v_i and v_j are different eigenvectors with respect to different eigenvalues, we can calculate:

$$v_j^T H v_i = (Hv_j)^T v_i = \lambda_j (Gv_j)^T v_i = \lambda_j v_j^T G(v_i) \quad \text{and} \quad v_j^T H v_i = v_j^T \lambda_i G(v_i) = \lambda_i v_j^T G(v_i)\,.$$

Hence we have the equality

$$\lambda_j (Gv_j)^T v_i = \lambda_i v_j^T G(v_i) \Leftrightarrow \underbrace{(\lambda_j - \lambda_i)}_{\neq 0} v_j^T G(v_i) = 0$$

Hence all eigenspaces are orthogonal and we can use Gram Schmitt to make the bases of the eigenspaces orthogonal and by rescaling orthonormal.

Let $L = [v_1|\dots|v_m]$ be the matrix whose columns are these G-orthonormal eigenvectors. The linear change of coordinates $y = Lz$ leads to new coordinates z. In these new coordinates, the quadratic forms transform as follows:

$$q_G(y) = y^T G y = (Lz)^T G(Lz) = z^T L^T G L z = z^T I z = \sum_{i=1}^{m} z_i^2$$
$$q_H(y) = y^T H y = (Lz)^T H(Lz) = z^T L^T H L z$$

Since $Hv_i = \lambda_i G v_i$, we have $L^T H L = \mathrm{diag}(\lambda_1, \dots, \lambda_m)$. The eigenvalues λ_i are real and have the same signature as the eigenvalues of H (l negative, $m - l$ positive).This is true to sysvesters law of inertia. By a further scaling of the v_i and a reordering the matrix $L^T H L$ can be brought into the form $\mathrm{diag}(-1, \dots, -1, 1, \dots, 1)$. (by doing this however, the matrix $L^T G L$ becomes $L^T G L = \mathrm{diag}(\mu_1, \dots \mu_m)$ with $\mu_j > 0 \ \forall j$). If ψ is the Morse chart we started with, then $L^{-1} \circ \psi$ is the new Morse chart:

$$f \circ (\psi^{-1} \circ L)(y) = f \circ \psi^{-1}(L(y)) = q_H(L(y) = \sum_{i=1}^{l} -y_i^2 + \sum_{i=l+1}^{m} y_i^1$$

Furthermore, we want to inspect the metric $g\big|_p : T_p M^2 \to \mathbb{R}$ with respect to the chart $L^{-1} \circ \psi$-induced basis $\left(\frac{\partial}{\partial x_1}\big|_p, \dots, \frac{\partial}{\partial x_m}\big|_p\right)$. By the lemma 2.0.9

$$g(q_{L^{-1}\circ\psi}\big|_p^{-1}, q_{L^{-1}\circ\psi}\big|_p^{-1})(v, w) = g(q_\psi\big|_p^{-1}, q_\psi\big|_p^{-1})(Lv, Lw) = (Lv)^T G L W = v^T L^T G L w$$

□

Definition 2.0.11. We call a Morse chart where the Matrix corresponding to the gradient in the critical point is of diagonal form a **refined Morse chart**.

Using such a chart, we can start looking at the flow itself and try to find easy local formulars around critical points. For this we start by looking at the Jacobian of the gradient in a critical point:

Definition 2.0.12 (Jacobian of the Gradient)**.** The gradient is a section into the tangent bundle, $\operatorname{grad}(f) : M \to TM$. Let $\psi : M \supseteq U \to V \subseteq \mathbb{R}^m$ be a chart. The coordinate map q_ψ on TU assigns to a vector $v \in T_pM$ (where $p \in U$ with $\psi(p) = x$) its components with respect to the basis $\left(\frac{\partial}{\partial x_i}\big|_p\right)$, i.e., $q_\psi(v) = (v_1, \dots, v_m)$ if $v = \sum_{i=1}^m v_i \frac{\partial}{\partial x_i}\big|_p$. We define the Jacobian of the gradient with respect to the chart ψ as the Jacobian matrix of the coordinate representation of the gradient in this chart:

$$J(\operatorname{grad}(f))_\psi(x) := J\left(q_\psi \circ \operatorname{grad}(f) \circ \psi^{-1}\right)(x) = \left(\frac{\partial (q_\psi \circ \operatorname{grad}(f) \circ \psi^{-1})_i}{\partial x_j}(x)\right)_{ij}$$

Lemma 2.0.13. *Let ψ be a coordinate system arround a critical point p. I.e. $\psi : p \in U \to V$ with $\psi(p) = 0$ and let $\left(\frac{\partial}{\partial x_1}\big|_x, \dots, \frac{\partial}{\partial x_m}\big|_x\right)$ be the induced basis of T_xM Then the Jacobian of the gradient reads*

$$J(\operatorname{grad}(f))_\psi(0) = \left(\sum_k g^{ki}(0) \frac{\partial^2 f \circ \psi^{-1}}{\partial x_j \partial x_k}\right)_{ij} . \tag{4}$$

Proof. Let q_ψ denote the coordinate funktion $TU \to R^m$ induced from the basis $\left(\frac{\partial}{\partial x_1}\big|_x, \dots, \frac{\partial}{\partial x_m}\big|_x\right)$. then the gradient in ψ reads:

$$q_\psi \circ \operatorname{grad}(f) = \left(\sum_k g^{k1} \frac{\partial f}{\partial x_k}, \dots, \sum_k g^{km} \frac{\partial f}{\partial x_k}\right)$$

Hence, we can differentiate:

$$\begin{aligned}
\frac{\partial}{\partial x} q_\psi \circ \operatorname{grad}(f) \circ \psi^{-1} &= \left(\frac{\partial}{\partial x_j} \sum_k g^{ki} \frac{\partial f \circ \psi^{-1}}{\partial x_k}\right)_{ij} \\
&= \left(\sum_k \left[\frac{\partial g^{ki}}{\partial x_j} \frac{\partial f \circ \psi^{-1}}{\partial x_k} + g^{ki} \frac{\partial^2 f \circ \psi}{\partial x_j \, \partial x_k}\right]\right)_{ij} .
\end{aligned}$$

and now in $p = \psi^{-1}(0)$ we have

$$\frac{\partial}{\partial x} q_\psi \circ \operatorname{grad}(f) \circ \psi^{-1}\Big|_0 = \left(\sum_k \left[g^{ki}(0) \frac{\partial^2 f \circ \psi^{-1}}{\partial x_j \, \partial x_k}\right]\right)_{ij} .$$

□

Now that we now how the gradient locally looks like ist reasonable to assume, that the flow locally looks like its exponential:

Lemma 2.0.14. *Let $t \in \mathbb{R}$ be small enough and $\varphi_t : M \to M$ be the flow map corresponding to the negative gradient. Assume that U is the domain of a chart ψ and $p \in U$ such that $\varphi_t(p) \in U$. Then the linear map $\mathrm{d}\varphi_t\big|_p : T_pM \to T_{\varphi_t(p)}M$ has a local representation in the coordinates induced by ψ given by:*

$$\frac{\partial}{\partial x}\psi \circ \varphi_t(\psi^{-1}(x)) = \exp\left(-\left(\sum_k g^{ki}(0)\frac{\partial^2 f \circ \psi^{-1}}{\partial x_j \partial x_k}\right)_{ij} \cdot t\right).$$

Proof. For this we consider the function $(t, x) \mapsto \psi \circ \varphi_t(\psi^{-1}(x)) : \mathbb{R} \times U \to U$ For fixed x we can calculate

$$q_\psi \frac{\mathrm{d}}{\mathrm{d}t}\varphi_t(\psi^{-1}(x)) = -q_\psi \circ \operatorname{grad}(f)(\varphi_t(\psi^{-1}(x)))$$

and by changing the order of integration we have:

$$\begin{aligned}
\frac{\partial}{\partial t}\frac{\partial}{\partial x}\psi \circ \varphi_t(\psi^{-1}(x)) &= \frac{\partial}{\partial x}\frac{\partial}{\partial t}\psi \circ \varphi_t(\psi^{-1}(x)) \\
&= \frac{\partial}{\partial x} q_\psi \frac{\mathrm{d}}{\mathrm{d}t}\varphi_t(\psi^{-1}(x)) \\
&= -\frac{\partial}{\partial x} q_\psi \circ \operatorname{grad}(f)(\varphi_t(\psi^{-1}(x))) \\
&= -\frac{\partial}{\partial x} q_\psi \circ \operatorname{grad}(f)\left(\psi^{-1} \circ \psi \circ \varphi_t(\psi^{-1}(x))\right) \\
&= -\frac{\partial}{\partial x}\left(q_\psi \circ \operatorname{grad}(f) \circ \psi^{-1}\right)\frac{\partial}{\partial x}\left(\psi \circ \varphi_t(\psi^{-1}(x))\right)
\end{aligned}$$

Hence by the theory of linear differential equation with $\frac{\partial}{\partial x}\psi \circ \varphi_0(\psi^{-1}(x)) = \mathbb{1}_m$ we have:

$$\frac{\partial}{\partial x}\psi \circ \varphi_t(\psi^{-1}(x)) = \exp\left(-\frac{\partial}{\partial x}\left(q_\psi \circ \operatorname{grad}(f) \circ \psi^{-1}\right) \cdot t\right).$$

The preceding lemma now finishes the proof.

□

Hence in a refind Morse chart we have:

Remark 2.0.15. Let ψ be a refined Morse chart, and hence:

(a) $g^{ik}(0) = \delta_{ik}\mu_k$ with $\mu_k > 0$ for any k.

(b) $\frac{\partial^2 f \circ \psi^{-1}}{\partial x_j \partial x_k} = s_k 2\delta_{jk}$ where $s_k = -1$ for $k \leq \lambda_p$ and $s_k = 1$ else.

Then the Jacobian of the flow in the critical point p reads:

$$\begin{aligned}\frac{\partial}{\partial x}\psi\circ\varphi_t(\psi^{-1}(x)) &= \exp(-J(\text{grad}(f))(p)\cdot t)\\ &= \exp\left(-\sum_k \underbrace{g^{ki}}_{=\delta_{ik}\mu_k}(0)\underbrace{\frac{\partial^2 f\circ\psi^{-1}}{\partial x_j\partial x_k}}_{=s_k 2\delta_{jk}}\cdot t\right)\\ &= \text{diag}\left(e^{2\mu_1 t},\dots,e^{2\mu_{\lambda_p}t},-e^{2\mu_{\lambda_p+1}t},\dots,-e^{2\mu_m t}\right)\end{aligned}$$

In a refined Morse chart we have that the (un-)stable Manifold embedding:

Definition 2.0.16 (Stable and Unstable Manifold)**.** Let $p\in M$ be a critical point of $f:M\to\mathbb{R}$. Then we define:

1. The **stable set** of p as
$$W(\to p):=\{x\in M|\lim_{t\to\infty}\varphi_t(x)=ß\}.$$
That is the points that flow towards p.
2. The **unstable set** of p as
$$W(p\to):=\{x\in M|\lim_{t\to-\infty}\varphi_t(x)=ß\}.$$
That is the points that came from p.

Those sets will turn out to be manifolds, which explains the names **(un-)stable manifolds**.

The main theorem of this chapter will be the:

Theorem 2.0.17 (Stable and unstable manifold theorem)**.** *Let $f:M\to\mathbb{R}$ be a smooth function on a smooth compact Riemannian manifold (M,g). Let p be a critical point of f. Then the tangent space splits as:*

$$T_pM = T_p^s(M)\bigoplus T_p^uM$$

such that the Hessian is positive definite on T_p^sM and negative definite on T_p^uM. Moreover, the stable and unstable manifolds are submanifolds and images of smooth embeddings:

$$\begin{aligned}E_p^s: T_p^sM &\to W(\to p)\subset M,\\ E_p^u: T_p^uM &\to W(p\to)\subset M.\end{aligned}$$

Proof. For the proof compare [BH04]. □

We want to refine the unstable embedding $E^u_q : T^u_q M \to W(q \to)$. For this we will make use of the Grobman-Hartman conjugation theorem:

Theorem 2.0.18 (Hartman-Grobman Theorem)**.** *Let E be an open subset of R^n cobtaining the origin, let $f \in \mathcal{C}$, and let φ_t be a flow given by f. (i.e. $\frac{\partial}{\partial t}(\varphi_t(x)) = f(x)$). Assume that $f(0) = 0$, i.e. that 0 is a fixpoint of φ_t and furthermore assume that fixpoint to be hyperbolic, i.e. Df has no eigenvalue with zero real part. Then there exists a homeomorphism H of an open set U containing the origin onto an open set V containing the origin such that for each $x_0 \in I$, there is an open Interval $I_0 \subseteq \mathbb{R}$ containing zeri such that for all $x_0 \in U$ and $t \in I_0$ the diagram*

$$\begin{tikzcd} U \arrow[r, "H"] \arrow[d, "\varphi_t"] & V \arrow[d, "\mathrm{e}^{At}"] \\ U \arrow[r, "H"] & V \end{tikzcd}$$

commutes ,meaning tha locally the flow is a topological conjugate to its linearisation.

Proof. We want to refine the unstable embedding $E^u_q : T^u_q M \to W(q \to)$. For this we will make use of the Grobman-Hartman conjugation theorem: Chapter 2.8 in [Per01]. □

Theorem 2.0.19 (Refined Stable Embdeding)**.** *In the setting of the stable manifold theorem, we can refine the embedding*

$$E^u_q : T^u_q M \to W(q \to)\,,$$

such that it resembles the flow, meaning that for $x = E^s_p(v)$ the flow is given by

$$\varphi_t(x) = E^s_p(A_t v)\,,$$

where A_t is a linear map defined as: $A_t = B_{\mathfrak{b}} \circ \operatorname{diag}\left(e^{-2\mu_1 t}, \ldots, e^{-2\mu_{\lambda_p} t}\right) \circ B_{\mathfrak{b}}^{-1}$, where $\mathfrak{b}$ is the bases of $T^u_q M$ that comes from the basis of $T_q M$ which is induced by a refined Morse chart. The diagonal matrix is the stable part of the jacobean of the flow in said refined Morse chart ψ.

Proof. In the proof of the unstable manifold theorem we started with a local map $\rho' : U \to V$, where $U \subseteq W(q \to)$ is an neighbourhood of q and $V \supseteq T^u_q$. (This was a refined

Morse chart together with a projection once realized that the stable part gets mapped to the graph of a smooth funktion). Now we define

$$\begin{aligned}\theta_t : V &\to V\\ v &\mapsto \rho' \circ \varphi_t \circ \rho'^{-1}(v)\end{aligned}$$

for all t such that the map is well defined. Now with a basis $\mathfrak{b}_p^s$ of $T_p^s M$ we can push these considerations into $\mathbb{R}^n$ and by abuse of notation assume that ρ maps to $\mathbb{R}^n$ and hence θ is a flow arround the origin of $\mathbb{R}^n$. Now by the Hartman-Grobman Theorem we find a homeomorphism H arround the origin, such that $H \circ \theta_t = e^{At} \circ H$, where $A = \mathrm{d}\theta_t =$ To calculate said differential we look into the map ρ' again. For this we give names to the funktions:

- $\psi : U \to V$ is the refined morse chart arround q.
- $g : T_p^s M \to T_p^u M$ is the map that defines the image of the stable part under ψ to be a graph. I.e. $\psi\left(W(\to p)\right) = \Gamma_g$. We know that $g(0) = 0$ and $\mathrm{d}g\big|_0 = 0$.
- $\pi : \Gamma_g \to T_p^s M$ is the projection. Its inverse is obviously $\mathrm{id} \times g$.

We know from the local stable manifold theorem, that $\mathrm{d}g\big|_0 = 0$. With this we can calculate:

$$\begin{aligned}\mathrm{d}\theta_t\big|_0 &= \mathrm{d}\left(\pi \circ \psi \circ \varphi_t \circ \psi^{-1} \circ \pi^{-1}\right)\big|_0\\ &= \mathrm{d}\pi\big|_{(0,0)} \circ \mathrm{d}\psi \circ \varphi_t \circ \psi^{-1}\big|_{0,0} \circ \mathrm{d}\,\mathrm{id} \times g\big|_0\\ &= \left(\mathrm{id}_{m-\lambda_p}, 0_{\lambda_p}\right) \circ \mathrm{diag}\left(e^{-2\mu_1 t}, \dots, e^{-2\mu_{\lambda_p} t}, e^{2\mu_{\lambda_q+1} t}, \dots, e^{2\mu_m t}\right) \circ \left(\mathrm{id}_{m-\lambda_p}, 0_{\lambda_p}\right)^T\\ &= \mathrm{diag}\left(e^{-2\mu_1 t}, \dots, e^{-2\mu_{\lambda_p} t}\right)\end{aligned}$$

Now define for a small neighbourhood $V \subseteq \mathbb{R}^{m-\lambda_p}$ the map

$$\rho := \rho' \circ H : V \to U\,.$$

U is choosen such that the map is onto. Then we have that $\varphi_t \circ \rho^{-1} = \rho^{-1} \circ \mathrm{e}^{At}$, in a that small neighbourhood of p. Now finally we can extend this property to be true for the whole stable set: For this we define the embedding

$$\begin{aligned}E_p^s : T_p^s M &\to W(\to p)\\ x &\mapsto \varphi_{-n} \circ \rho^{-1} \circ A_n(x) \quad \text{where } n \in \mathbb{N} \text{ is big enough such that } A_n(x) \in V\,.\end{aligned}$$

This map is in fact well defined since

$$\begin{aligned}\varphi_n \circ \rho^{-1} \circ A_n(x) &= \varphi_{-n} \circ \varphi_{-1} \circ \varphi_1 \circ \rho^{-1} \circ A_n(x)\\ &= \varphi_{-n} \circ \varphi_{-1} \circ \rho^{-1} \circ A_{n+1}(x)\\ &= \varphi_{-(n+1)} \circ \rho^{-1} \circ A_{n+1}(x)\end{aligned}$$

And obviously the flow compatibility that is given locally by ρ is now true globally: Let $x \in W(\to p)$ and assume $x = E_p^s(y)$. Then:

$$\begin{aligned}\varphi_t(x) = \varphi_t(E_p^s(y)) &= \varphi_t \circ \varphi_{-n} \circ \rho^{-1} \circ A_n(y)\\ &= \varphi_{-n} \circ \varphi_t \circ \rho^{-1} \circ A_n(y)\\ &= \varphi_{-n} \circ \circ A_n(A_t(y))\end{aligned}$$

□

Remark 2.0.20. Notice how the sign of the determinant of A_t is always positiv.

Now we want to understand the flow in an unstable set and analyze two maps:

Remark 2.0.21. The Map

$$\tau : W(q \to) \setminus \{q\} \to \mathbb{R} \tag{5}$$

$$x \mapsto \tau(x) \text{ such that } f \circ \varphi_{\tau(x)}(x) = c \tag{6}$$

is smooth.

Proof. We know that such a map exists and is well defined since every flow line passes through $f^{-1}(c)$. Now let $x \in W(q \to) \setminus \{q\}$, V be a neighbourhood of x and U be a neighbourhood of $\varphi_{\tau(x)}(x)$. Define the map

$$\begin{aligned}G : V \times I &\to \mathbb{R}\\ (x,t) &\mapsto f \circ (\varphi_t(x)) \,.\end{aligned}$$

Now the partial derivative reads

$$\frac{\partial}{\partial t} G\big|x,t = \mathrm{d}f\big|_{\varphi_t(x)} \circ \frac{\partial}{\partial t}\varphi_t(x)\big|_t = \mathrm{d}f\big|_{\varphi_t(x)} \circ \left(-\mathrm{grad}_{\varphi_t(x)}(f)\right) .$$

For $\varphi_{\tau(x)}$ we have that $T_{\varphi_{\tau(x)}(x)} f^{-1}(c) = \ker \mathrm{d}f\big|_{\varphi_{\tau(x)}(x)}$. Since the flow line is transversal to the level set or said differently, the gradient is orthogonal to level sets, we have that the latter does not live in the kernal and hence

$$\frac{\partial}{\partial t} G\big|x,\tau(x) = \mathrm{d}f\big|_{\varphi_{\tau(x)}(x)} \left(-\mathrm{grad}_{\varphi_{\tau(x)}(x)}(f)\right) \neq 0 \,.$$

This lets us use the implicit function theorem to deduce that the level set $G = c$ is ocally the graph of a funktino $\tau : U_x \to \mathbb{R}$ where U_x is a neighbourhood of x. Hence the function τ is globally smooth since this argument can be done everywhere. □

Remark 2.0.22. Now we care for the differential of τ in a point $x_j \in W(q \to) \cap f^{-1}(c)$. For this we choose a chart of $W(q \to)$ arround x_j such that the induced bases of $T_{x_j} W(q \to) = \langle -\mathrm{grad}_{x_j}(f) \rangle \oplus T_{x_j} V_j$ that respects the splitting. for example $\frac{\partial}{\partial x^1}\big|_{x_j} =$

$-\text{grad}_{x_j}(f)$ and $\frac{\partial}{\partial x^i}\big|_{x_j} \in T_{x_j}V_j$ for i bigger than one. With this we can start our calculations: Notice how $x \mapsto f \circ \varphi_{\tau(x)}(x)$ is a constant function and hence its differential is 0.

$$\begin{aligned} 0 = \mathrm{d}_x\left(f \circ \varphi_\tau\right) = \overbrace{\mathrm{d}f}^{\text{1-form}} \left(\mathrm{d}\varphi_\tau\right) &= \mathrm{d}f\left(\mathrm{d}_t\varphi_t \circ \mathrm{d}_x\tau(x) + \mathrm{d}_x\varphi_{\tilde{t}=\tau(x)}\right) \\ &= \mathrm{d}f\left(\mathrm{d}_t\varphi_t \circ \mathrm{d}_x\tau(x)\right) + \mathrm{d}f\left(\mathrm{d}_x\varphi_{\tilde{t}=\tau(x)}\right) \\ &= \mathrm{d}_t\left(f \circ \varphi_t\right) \circ \mathrm{d}_x\tau(x) \circ + \mathrm{d}f\left(\mathrm{d}_x\varphi_{\tilde{t}=\tau(x)}\right) \end{aligned}$$

Hence we know that

$$\mathrm{d}_x\tau = -\frac{\mathrm{d}f\left(\mathrm{d}_x\varphi_{\tilde{t}=\tau(x)}\right)}{\mathrm{d}_t\left(f \circ \varphi_t\right)} = -\frac{\mathrm{d}f\left(\mathrm{d}_x\varphi_{\tilde{t}=\tau(x)}\right)}{-\left\|\text{grad}(f)^2\right\|} \tag{7}$$

Now we can calculate in x_j where $\tau(x_j) = 0$

$$\mathrm{d}_x\tau\big|_{x_j} \frac{\partial}{\partial x^i}\Big|_{x_j} = \frac{\mathrm{d}f\big|_{x_j}\frac{\partial}{\partial x^i}\big|_{x_j}}{\left\|\text{grad}(f)^2\right\|} = \frac{g\left(\text{grad}_{x_i}(f), \frac{\partial}{\partial x^1}\big|_{x_j}\right)}{\left\|\text{grad}(f)^2\right\|} = \begin{cases} -1 & \text{if } i = 1 \\ 0 & \text{else.} \end{cases}$$

This result can be extended to also work outside of x_j as follows: Notice the identity for all $s \in \mathbb{R}$ and $x \in W(q \to) \setminus \{q\}$:

$$\tau(\varphi_s(x)) = \tau(x) - s\,.$$

Differentiating this in the variable s and in for s' yealds

$$\mathrm{d}\tau\big|_{\varphi_{s'}(x)} \circ \underbrace{\mathrm{d}\varphi_s(x)\big|_{s=s'}}_{=-\text{grad}_{\varphi_s(x)}(f)} = \mathrm{d}\tau(\varphi_{s'}(x))\big|_{s'} = \mathrm{d}\tau(x) - s\big|_{s'} = -1\,.$$

And hence in $s' = 0$ we have for all $x \in W(q \to) \setminus \{q\}$:

$$\mathrm{d}\tau\big|_x\left(-\text{grad}(x)(f)\right) = -1\,.$$

We can furthermore define a basis of $T_yW(q \to)$ with $\mathfrak{b}_1 := -\text{grad}_y(f)$ and $\mathfrak{b}_i = \mathrm{d}\varphi_{\tau(y)}\big|_y^{-1}(w_i)$ where (w_i) denotes a basis of $T_{x_j}V_j$. Hence w_i is orthogonal on $-\text{grad}_{x_j}(f)$ for all $i > 1$. Hence by the equation (7) we can caluclate:

$$\mathrm{d}_x\tau(\mathfrak{b}_i) = -\frac{\mathrm{d}f\left(\mathrm{d}_x\varphi_{\tilde{t}=\tau(x)}(\mathfrak{b}_i)\right)}{-\left\|\text{grad}(f)^2\right\|} = -\frac{\mathrm{d}f\left(w_i\right)}{-\left\|\text{grad}(f)^2\right\|} = 0 \quad \text{for all } i > 1\,.$$

Remark 2.0.23. The map

$$\begin{aligned} \rho : W(q \to) \setminus \{q\} &\to W(q \to) \cap f^{-1}(c) \\ x &\mapsto \varphi_{\tau(x)}(x)\,. \end{aligned}$$

is a submersion in $y \in S(q \to)$.

Proof. To prove this we calculate its differential $\mathrm{d}\rho\big|_y$ as follows:

$$\mathrm{d}\rho\big|_y = \mathrm{d}_x(\varphi_\tau)\big|_y = \mathrm{d}_t\varphi_t\big|_{t=\tau(x)} \circ \mathrm{d}_x\tau\big|_y + \mathrm{d}_x\varphi_{\tau(y)}\big|_y$$

We can rewrite this by viewing $\mathrm{d}_x\tau\big|_y$ as a one form and using the definition of the flow: $\mathrm{d}_t\varphi_t\big|_{t=\tau(x)} = -\mathrm{grad}_{\rho(x)}(f)$:

$$\mathrm{d}\rho\big|_y = -\mathrm{grad}_{\rho(x)}(f) \cdot \mathrm{d}_x\tau\big|_y + \mathrm{d}_x\varphi_{\tau(y)}\big|_y$$

Now if $y \in f^{-1}(c)$ we have that $\tau(y) = 0$ and hence $\mathrm{d}_x\varphi_{\tau(y)} = \mathrm{id}$. Hence the differential reads:

$$\mathrm{d}\rho\big|_y = -\mathrm{grad}_y(f) \cdot \mathrm{d}_x\tau\big|_y + \mathrm{id}\ .$$

Now we can choose a local chart arround y such that $-\mathrm{grad}_y(f)$ becomes the first basis vector of the tangend space, and the rest are orthonormal. Then the differential acts on $-\mathrm{grad}_y(f)$:

$$\mathrm{d}\rho\big|_y(-\mathrm{grad}_y(f)) = -\mathrm{grad}_y(f) \cdot \underbrace{\mathrm{d}_x\tau\big|_y(-\mathrm{grad}_y(f))}_{=-1} - \mathrm{grad}_y(f) = 0$$

and on all the other tangend vectors as the identity, due to the first summand vanishing. Therefore the jacobean in this basis is of the form

$$\mathrm{d}\rho\big|_y = (0, 1_{\lambda_q - 1}).$$

□

Lemma 2.0.24. *Assume that $\mathfrak{d} = (\mathfrak{d}_i)$ is a basis of $T^u_q M$. Let $y := E^{u\,-1}_q(x_j)$ and $y' \in (\rho \circ E^u_q)^{-1}(x_j)$. Then the map*

$$\mathrm{d}E^u_q\big|_y^{-1} \circ \mathrm{d}\rho \circ \mathrm{d}E^u_q\big|_{y'} \circ Q_{y'} \circ B_{\mathfrak{d}} : \mathbb{R}^{\lambda_q} \to T_y T^u_q M$$

can be written as

$$\begin{aligned} \pi_{-\mathrm{grad}_{x_j}(f)^\top} \circ (B_{\mathfrak{b}^q_{x_j}}) : \mathbb{R}^{\lambda_q} &\to T_{x_j} V_j \\ e_i &\mapsto \begin{cases} 0 & \text{for } i = 1 \\ (\mathfrak{b}^q_{x_j})_i & \text{for } i \neq 1\,. \end{cases} \end{aligned}$$

where $(\mathfrak{b}^q_{x_j})_j$ is basis of $T_{x_j} W(q \to)$ that is induced from $\mathfrak{d}$ and $(\mathfrak{b}^q_{x_j})_i = -\mathrm{grad}_{x_j}(f)$. In other words, this is a basis compatible with the one used in the counting of flow lines in the Morse boundary operator.

Proof Strategy:
To proove this, we first remined ourself how the basis $(\mathfrak{b}^q_{x_j})_{j>1}$ is constructed. It is a basis that when extendet with $-\mathrm{grad}_{x_j}(f)$ as the first vector is compatible with $\mathfrak{d}$. Hence, we can construct it from $\mathfrak{d}$, by assuming that $\mathrm{d}E^u_q\big|_y \circ Q_y \circ \mathfrak{d}_1 = -\mathrm{grad}_y(f)$. Then, if we push $\mathfrak{d}$ to $T_{x_j}W(q \to)$ via $\mathrm{d}E^u_q\big|_y \circ Q_y$ and delete the first vector, we get the Basis we wanted. So what we need to check in order to proof this lemma is, if the maps

$$\begin{aligned} \mathrm{d}E^u_q\big|_y \circ Q_y \circ B_{\mathfrak{d}} : \mathbb{R}^{\lambda_q - 1} &\to \left(-\mathrm{grad}_{x_j}(f)\right)^{\top} \\ \mathrm{d}E^u_q\big|_y^{-1} \circ \mathrm{d}\rho \circ \mathrm{d}E^u_q\big|_{y'} \circ Q_{y'} \circ B_{\mathfrak{d}} : \mathbb{R}^{\lambda_q - 1} &\to \left(-\mathrm{grad}_{x_j}(f)\right)^{\top} \end{aligned}$$

induce the same orientation when restricted to the same domain such that they become vector space isomorphisms. To proof this we will work from the bottom up: We adapt the basis $\mathfrak{d}$ to work with the lower map and show, later, that the adapted basis fits the requirements above:

Proof. We start by defining the space $\left(\tau \circ E^u_q\right)^{-1}(\tilde{t})$, where $\tilde{t} := \tau \circ E^u_q(y')$. As a level set, this is a submanifold of $T^u_q M$ and its tangend space in y' is given by the basis $E^u_q\big|_{y'}^{-1}(\mathfrak{b}_i)$ for $i > 1$, where $\mathfrak{b}_i = \mathrm{d}\varphi_{\tilde{t}}\big|_{y'}^{-1}(w_i)$ where (w_i) denotes a basis of $T_{x_j}V_j$. Now we have the equality of maps:

$$A_{\tilde{t}}\big|_{\left(\tau \circ E^u_q\right)^{-1}(\tilde{t})} = \left(E^{u-1}_q \circ \rho \circ E^u_q\right)\big|_{\left(\tau \circ E^u_q\right)^{-1}(\tilde{t})}$$

Hence for a basis $\mathfrak{d}_i$ of $T^u_q M$ such that $\mathrm{d}E^u_q\big|_{y'} \circ Q_{y'}(\mathfrak{d}_i)$ is $-\mathrm{grad}_{y'}(f)$ for $i = 1$ and $\mathfrak{b}_i$ else, we have:

$$\mathrm{d}\left(E^{u-1}_q \circ \rho \circ E^u_q\right)\big|_{y'} \circ Q_{y'}(\mathfrak{d}_i) = Q_y \circ A_{\tilde{t}}(\mathfrak{d}_i).$$

For $i = 1$, the left side maps to 0, and the right side:

$$\begin{aligned} Q_y \circ A_{\tilde{t}}(\mathfrak{d}_1) &= \mathrm{d}A_{\tilde{t}}\big|_{y'} \circ Q_{y'}(\mathfrak{d}_1) \\ &= \mathrm{d}A_{\tilde{t}}\big|_{y'} \circ \mathrm{d}E^u_q\big|_{y'}^{-1}(-\mathrm{grad}_{y'}(f)) \\ &= \mathrm{d}A_{\tilde{t}} \circ \mathrm{d}E^u_q\big|_{y'}^{-1}(-\mathrm{grad}_{y'}(f)) \\ &= \mathrm{d}(E^u_q)^{-1} \circ \varphi_{\tilde{t}}\big|_{y'}^{-1}(-\mathrm{grad}_{y'}(f)) \\ &= \mathrm{d}E^u_q\big|_{x_j}^{-1} \circ \mathrm{d}\varphi_{\tilde{t}}\big|_{y'}^{-1}(-\mathrm{grad}_{y'}(f)) \\ &= \mathrm{d}E^u_q\big|_{x_j}^{-1}\left(-\mathrm{grad}_{x_j}(f)\right) \end{aligned}$$

The last equation is the general fact prooven in remark 2.0.15, that for fixed t the identity holds:

$$\mathrm{d}_x\varphi_t\big|_y(-\mathrm{grad}_y(f)) = -\mathrm{grad}_{\varphi_t(y)}(f)$$

Hence if π denotes the projection onto the complement of $\mathrm{d}E_q^u\big|_{x_j}^{-1}\left(-\mathrm{grad}_{x_j}(f)\right)$ we have an equality of maps

$$\mathrm{d}\left(E_q^{u-1}\circ\rho\circ E_q^u\right)\Big|_{y'}\circ Q_{y'}=\pi\circ Q_y\circ A_{\tilde{l}}.$$

Or by the above calculation, we can throw out π, if we exclude $\mathfrak{d}_1$. Furthermore, the equation shows that our requirement $\mathrm{d}E_q^u\big|_{y'}\circ Q_{y'}(\mathfrak{d}_1)=-\mathrm{grad}_{y'}(f)$ implies that $\mathfrak{d}$ is compatible with a basis where $\mathrm{d}E_q^u\big|_y\circ Q_y\circ\mathfrak{d}_1=-\mathrm{grad}_y(f)$. This concludes the prove! □

Definition 2.0.25 (Morse Homology)**.** Let $f:M\to\mathbb{R}$ be a Morse-Smale function on a Riemannian manifold (M,g). Furthermore let $\mathfrak{o}$ be an choice of orientations for all unstable Tangendspaces (T_q^uM for all $q\in\mathrm{Crit}(f)$). The tuple $\mathfrak{A}:=(g,f,\mathfrak{o})$ is a **Morse datum**. With this we define now the **Morse chain groups** for a given commutative ring with one (Λ):

$$C_k(M,\mathfrak{A},\Lambda):=\bigoplus_{\mathrm{Crit}(f)}\Lambda$$

To define a boundary operator between the chain groups on the generators we count flow lines between them. Lets say that p is a critical point of index k and q one of index $k+1$. Then we have the set of flow lines from q to p denoted by $\mathcal{M}(q\to p)$. For each $\gamma\in\mathcal{M}(q\to p)$ we assign a sign $n_\gamma\in\{\pm1\}$. Let $x\in\gamma$. Then the tangent space in x splits $T_xW(q\to)=N_x(\gamma)\oplus\langle-\mathrm{grad}(f)(x)\rangle$. Furthermore, we can parallel transport $N_x(\gamma)$ to T_p^uM. Hence, the orientation in T_q^uM can be compared to the one in T_p^uM and the sign n_γ can be chosen accordingly.The **boundary operator** on a generator is defined as:

$$\partial_{k+1}(q):=\sum_{p\in\mathrm{Crit}_k(f)}\sum_{\gamma\in\mathcal{M}(q\to p)}n_\gamma$$

This gives rise to a chain complex and hence gives rise to a homology theory, and a cohomology theory by applying the hom functor to the chain complex.

Definition 2.0.26 (Orientation and Exact Sequences)**.** As stated before, an orientation of a Vector space V of dimension m is given by an ordered basis $\mathfrak{b}=(b_1,\dots b_m)$ of V and hence the produkt $b_1\wedge\dots\wedge b_m$ is an element of the maximal exterior product $\det(V)=\Lambda^mV$. Two ordered bases define different elements in $\det(V)$ where there difference is given by the determinant of the base change matrix. (This is just a calculation using multilinearity and the alternating property). Hence an orientation is a choice of one connected component of $\det(V)\smallsetminus\{0\}$. Now a short exact sequence of vector spaces

$$0\longrightarrow A\xrightarrow{f}B\xrightarrow{g}C\longrightarrow0$$

induces a canonical isomorphism

$$\begin{aligned}\det(A)\otimes\det(C)&\to\det(B)\\(a_1\wedge\dots\wedge a_l)\otimes(c_1\wedge\dots\wedge c_k)&\mapsto(f(a_1)\wedge\dots f(a_l)\wedge\tilde{c_1}\wedge\dots\wedge\tilde{c_k}),\end{aligned}$$

where $\tilde{c_j}\in g^{-1}(c_j)$. The well definition of this map (invariance under choosen lift of c) can be calculated, since two different choices differ in an element in the kernal of g which equals the image of f and hence the different choices vanisch in the wedge produkt. By this isomorphism we get a orientation induced in the middle from the outside.

Remark 2.0.27 (The Counting of Flow Lines)**.** There are several ways to formulate the counting of flow lines that happens in the boundary operator.Let $\gamma \in \mathcal{M}(q \to p)$ and $x \in \gamma$ Instead of using the parallel transport, we can introduce two different ordered bases in $T_x W(q \to)$. We use the following notation: Let $\gamma \in \mathcal{M}(q \to p)$ and $x \in \gamma$. For this we define $N_x^{W(q\to)}\gamma$ be the normal bundle of γ as a submanifold of $W(q \to)$. Since γ is contractible we can trivialize said normal bundle to get an bundle isomorphism:

$$\nu : N^{W(q\to)}\gamma \to \gamma \times T_p^u M\,.$$

Let $\nu\big|_x$ denote the isomorphism $N_x^{W(q\to)}\gamma \to T_p^u M$. An orientation of $T_p^u M$ induces then one on $N_x^{W(q\to)}\gamma$ and if we define the bases $(-\mathrm{grad}_x(f))$ of its own span to be positive, we get the exact sequence

$$\begin{tikzcd} 0 \arrow[r] & \langle -\mathrm{grad}_x(f) \rangle \arrow[r] & T_x W(q\to) \arrow[r] & N_x^{W(q\to)}\gamma \arrow[r] & 0 \\ & & & T_p^u M \arrow[u, "(\nu_x)^{-1}"'] & \end{tikzcd}$$

That introduces an orientation in the middle that can be represented by the basis $(-\mathrm{grad}_x(f), i \circ \nu|_x)(\mathfrak{b}^p)$, where i is the inclusion $N_x^{W(q\to)}\gamma \to T_x W(q \to)$ and $(\mathfrak{b}^p)$ is a ordered basis ot $T_p^u M$. Another way to get an orientation there is to trivialize the bundle $TW(q \to)$ via the isomorphism:

$$\mu : TW(q \to) \to W(q \to) \times T_q^u M\,; \xi_y \mapsto \left(y, Q^{-1}_{E_q^{u-1}(y)} \circ \mathrm{d}E_q^{u-1}\big|_y (\xi_y)\right).$$

This trivialization together with an orientation on $T_q^u M$ orients the middle in the above exact sequence. Now the deciding factor that gives γ its sign is weather the two orientations agree. This can by checked by calculating the determinant of a mcertain map. To formulate the map somewhat readible, we assume that $\mathfrak{b}^q$ is a basis of $T_q^u M$ such that for $y = E_q^{u-1}(x)$ we have $\mathrm{d}E_q^u\big|_y \circ Q_y(\mathfrak{b}_i^q) = -\mathrm{grad}_x(f)$ for $i = 1$, and for all $i > 1$ the image lives in the orthogonal complement of $-\mathrm{grad}_x(f)$ and hence in the domain of ν_x. Then we can define the map

$$\begin{aligned} B^{-1}_{\mathfrak{b}^p} \circ \nu_x \circ \mu_x^{-1} \circ B_q : \mathbb{R}^{\lambda_q} \supseteq \mathbb{R}^{\lambda_q - 1} = \langle e_2, \dots e_{\lambda_q} \rangle &\to \mathbb{R}^{\lambda_q - 1} \\ e_i &\mapsto B^{-1}_{\mathfrak{b}^p} \circ \nu_x \circ \mathrm{d}E_q^u\big|_y \circ Q_y(\mathfrak{b}_i^q) \end{aligned}$$

Remark 2.0.28. Notice how we already saw a different way to calculate the sign by the map from lemma 2.0.24.

Here i write a little bit about the Morse cohomology in german

Theorem 2.0.29 (Morse Cohomology)**.** *Um die Morse Kohomologie zu erhalten, dualizieren wir den Kettenkomplex für einen Ring G mit 1. Sei* $\mathrm{Crit}_k = p_1, ...p_l$ *eine Basis der* $k-$*ten Kettengruppe. Für*

$$\begin{aligned} \eta_i : C_k &\to G \\ p_j &\mapsto \begin{cases} 0 & \textit{if } j = i \\ 1 & \textit{else} \end{cases} \end{aligned}$$

bekommen wir eine Basis $(\eta_1, ..., \eta_l)$ *der Cocetten Gruppe. Die Randabbilsung ist nun der spannende Teil. Seien* $(q_1, ..., q_r)$ *kritische Punkte von Index* $k+1$ *und* $(\xi_j)_{j=1,...,r}$ *die resultierende Basis von*C^{k+1}*. Dann*

$$\begin{aligned} \mathrm{d}_k : C^k &\to C^{k+1} \\ \eta_i &\mapsto \mathrm{d}\eta_i = \eta_i \circ \partial_{k+1} = \sum_j n_f(q_j, p_i)\xi_j. \end{aligned}$$

Hier beschreibt $n_f(q,p)$ *das zählen von Flusslionien von* q *nach* p *mittels* f*.Indem wir kritische Punkte und die formen identifizieren erhalten wird das folgende:*

$$\begin{aligned} C^k &= \bigoplus_{q \in \mathrm{Crit}_k} \mathbb{Z} = C_k. \\ \mathrm{d} :\ C^k &\to C^{k+1} \\ q &\mapsto \sum_{p \in \mathrm{Crit}_{k+1}} n_f(q,p) = \sum_{p \in \mathrm{Crit}_{k+1}} n_{-f}(p,q) = \partial_{-f}. \end{aligned}$$

Hier bezeichnet ∂_{-f} *den Randoperator* $C(M, -f, g)$*.*

Theorem 2.0.30 (Poincaré Dualität)**.** *Schnell sehen wir, dass*

$$C^k(M,f) = C_{m-k}(M,-f)$$

damit erhalten wir:

$$H^k(M) \cong H^k(M,f) = \ker(\partial_{-f}) \big/ \mathrm{im}\partial_{-f} = H_{m-k}(M,-f) \cong H_k(M)$$

3 What are Spheres

In this section we want to inspect representations of homotopical Spheres. To give a little overview we will define the sphere to come with an orientation. To be more specific we have the definition:

Definition 3.0.1 (The $k-$-Sphere)**.** Let $\mathbb{R}^k$ be the k-dimensional vector space given as the span of $e_1, \dots e_k$ together with the eucliden metric $\|\cdot\|_2$. Then the set

$$S^k := \{x \in \mathbb{R}^k | \, \|x\|_2 = 1\}$$

is called the $k-$**sphere**.

This sphere will be our save haven where we will always come back to. In this definition there is no room for "diffrent orientations on a sphere". In our reality we will detect the two "different orientations" of a space A homotopic to a sphere that arise from two constructions as follows: assume that α and β are the maps $A \to S^k$. then there is an endomorphism $f : S^k \to S^k$ such that our diagramm commutes:

$$\begin{tikzcd} & A \arrow[dl, "\alpha"'] \arrow[dr, "\beta"] & \\ S^k \arrow[rr, "f"] & & S^k \end{tikzcd}$$

Now $f^* : K^n(S^k) \to K^n(S^k)$ is an isomorphism and hence either the identity or minus the identity.

Now we have a lot of spaces that are homotopic or even homeomorph to the sphere. Just to name a view constructions:

1. The disc in $\mathbb{R}^k$ modulo its boundary $D^k/\partial D^k$.
2. The cone over the boundary of a D^k disc.
3. The resduced and unreduced susopension of a $k-1$ sphere.
4. The one-point compactification of a vector space together with an oriented basis.

We already saw how we can map the one-point compactification of a vector space with an ordered basis to a sphere wia the stereographic projections from the south pole. So lets inspect the other construction

Corollary 3.0.2. Given the disk $D^k \subseteq \mathbb{R}^k$ we have a homeomorphism

$$\overline{D^k}/\partial D^k \to S^k$$

Proof. We will construct a proper map from the open disk to R^k and by this we get a homeomorphism between their one-point-compaktifications.

$$\begin{aligned} s : D^k &\to \mathbb{R}^k \\ x &\mapsto x \cdot \frac{1}{1 - \|x\|_2} \end{aligned}$$

This is a homeomorphism and proper. Hence, we get a map between the onepoint compactifications

$$\overline{D^k}/\partial D^k \to \mathbb{R}^k \cup \{\infty\}$$

and the latter is homeomorphic to the Sphere. □

Corollary 3.0.3. Let $\overline{D^k} \cup C\partial D^k$ be the cone over the boundary of the disk. This is homeomorphic to the k–sphere.

Proof. Again we will use the language of onepoint compactifications. First we give a homeomorphism from the "open cone" over the boundary to the open disc. To do this we include our space into R^k as follows:

$$\begin{aligned} f : \overline{D^k} \cup \partial D^k \times [0,1) &\to R^k \\ x &\mapsto \begin{cases} x & \text{if } x \in \overline{D^k}\,, \\ (1+t)x & \text{if } (x,t) \in \partial D^k \times [0,1)\,. \end{cases} \end{aligned}$$

This is a homeomorphism (if we restrict the image to $\{x \in R^k | \, \|x\|_2 < 2\}$). This can be seen as follows: Obviously it is bijective and the contiuouty can be derived from the gluing lemma. To see the contiuoity of the invese we can explicitly describe it:

$$\begin{aligned} f^{-1} : \{x \in R^k | \, \|x\|_2 < 2\} &\to \overline{D^k} \cup \partial D^k \times [0,1) \\ x &\mapsto \begin{cases} x & \text{if } x \in \overline{D^k}\,, \\ (\frac{x}{\|x\|_2}, (1 - \|x\|_2)) & \text{if } \|x\|_2 \geq 1\,, \end{cases} \end{aligned}$$

Again, gluing lemma gives continouity, and a calculating gives us that they are inverse to each other. To see that the map is proper, we check if preimages of compact sets are compact. So let $K \subset \{x \in R^k | \, \|x\|_2 < 2\}$ be compact. Then $K = A \cup B$ where $A = K \cap \overline{D^k}$ and $B = K \cap \{x \in R^k | 1 \leq \|x\|_2 < 2\}$. Now A and B are again compact. (A is the intersection of two compact Hausdorff spaces and B can also be constructed as such a intersection) Since $f^{-1}(A \cup B) = f^{-1}(A) \cup f^{-1}(B)$ we have to check if f^{-1} is a closed map restricted to $\overline{D^k}$ and $\{x \in R^k | 1 \leq \|x\|_2 < 2\}$. But those restrictions give rise to homeomorphisms and hence they are closed. Now we have a proper homeomorphism which lets us conclude, that the map induced on their onepoint compactifications are also homeomorphisms. After the rescaling:

$$\begin{aligned} r : \{x \in R^k | \, \|x\|_2 < 2\} &\to D^k \\ x &\mapsto \frac{x}{2} \end{aligned}$$

we have a proper homeomorphism $\overline{D^k} \cup \partial D^k \times [0,1) \to D^k$. This lets us deduce the statment, as the onepoint compactification of the disk can be mapped to the sphere as seen in the corollary above. □

We will later use similar maps in our main prove, but now we want to talk about spheres in K−theorie.

Definition 3.0.4. Let $f : S^k \to S^k$ be continouos. We know that $f^* : K^k[S^k] \to K^k[S^k]$ is a group homomorphism and via the map $\mathbb{Z} \to K^k[S^k]$ induced from mapping $1 \mapsto \beta$ we get a induced map $\tilde{f}$ such that the diagramm kommutes:

$$\begin{tikzcd} K^k[S^k] \arrow[r, "f^*"] & K^k[S^k] \\ \mathbb{Z} \arrow[u] \arrow[r, "\tilde{f}"] & \mathbb{Z} \arrow[u] \end{tikzcd}$$

Hence $\tilde{f}$ is the multiplication with a number $d \in \mathbb{Z}$. This number is obviously homotopie invariant and hence we can define the degree map:

$$\begin{aligned} \deg : \pi_k(S^k) &\to \mathbb{Z} \\ [f] &\mapsto d \end{aligned}$$

Corollary 3.0.5. The map deg is an group isomorphism.

Proof. We start by prooving that it is a group homomorphism. Let f, g be representatives of two elements in $\pi_k(S^k)$. Let $\deg(f) = m$, and $\deg(g) = n$. The map $f + g$ is defined as the composition:

$$S^k \xrightarrow{c} S^k \wedge S^k \xrightarrow{f \wedge g} S^k$$

Here c denotes the collaps of the equator. □

Corollary 3.0.6. We want to understand homöomorphisms between boukettes of spheres. Or to beginn with let:

$$f : S^k \to \bigvee S^k$$

Now how does $f*$ look like? for this we

Proof. L □

Corollary 3.0.7. If we collaps a cone $C(S^k)$ over a shpere at its base, we obtain a sphere.

Proof. We define the homeomorphism as follows:

$$\begin{aligned} \Omega : C(S^k) \big/ \{0\} \times S^k &\to S^{k+1} \subset \mathbb{R}^{k+2} \\ \overline{(t, x)} &\mapsto (-\cos(\pi t), \sin(\pi t)x) \, . \end{aligned}$$

This homeomorphism has the following properties: If $t > \frac{1}{2}$ we end up in the upper hemissphere. Furthermore, we can view the domain and its image as topological manifolds and if we cut out the collapsed points, we even have a smooth manifold. Here the map is differentiable. □

We can use this map to construt a new map form $\overline{D^k}\big/\partial D^k \to S^k$ as follows:

Definition 3.0.8. We can identify $\overline{D^k}\big/\partial D^k$ with $C(S^{k-1})\big/\{0\} \times S^{k-1}$ as follows:

$$s : \overline{D^k}\big/\partial D^k \to C(S^{k-1})\big/\{0\} \times S^{k-1}$$
$$\overline{x} \mapsto \overline{(\|x\|, \frac{x}{\|x\|})}.$$

The composition $\Gamma := \Omega \circ s$ now gives a map

$$\overline{D^k}\big/\partial D^k \to S^k$$

Now we want to talk abpout charts of the Sphere. Assume that the sphere is opriented such that the projection of the upper hemissphere is an oriented chart. Now we want to construct a chart adapted to a Sphere that is given by $S^{k+1} = S \wedge D^k\Big/\partial D^k = \left(I \times D^k\right)\Big/\left(\partial(I \times D^k)\right)$. We have this chart as follows:

$$(\Omega \circ (id \wedge \Gamma))^{-1} : S^{k+1} \smallsetminus \{N\} \to \operatorname{int}(I \times D^k) \quad N \text{ denotes the north pole}.$$

, where $(\Omega \circ \mathrm{id} \wedge \Gamma))$ is given by

$$(t, x) \mapsto \left(-\cos(\pi t), \sin(\pi t)\Big(-cos(\pi \|x\|), sin(\pi(\|x\|) \frac{x}{\|x\|})\Big)\right)$$

Lemma 3.0.9. *Let $S^k = B^k\big/\partial B^k$. Let furthermore $D^k \subseteq B^k$ be a closed disk in B^k. Let $f : B^k \to D^k$ be the smooth diffeomorphism (that makes D^k a Disk). Let furthermore $A := \operatorname{int}(B^k) \smallsetminus f^{-1}(\partial D^k)$. Then $f|_A$ is a smooth map between open sets of $\mathbb{R}^k$ and hence we have a well defined jaccobean. We define $\bar{f} : B^k\big/\partial B^k \to D^k\big/\partial D^k$ to be the quotient map of f, and let $g : B^k\big/\partial B^k \to D^k\big/\partial D^k$ be the collapsing of the komplement of D^k. Then*

$$(\overline{f}^{-1} \circ g)^* : K^k[S^k] \to K^k[S^k]$$

is a group isomorphism and hence $\pm\,\mathrm{id}$. We have the equivalence of $(f^{-1} \circ g)^* = \mathrm{id}$ if and only if for any $x \in A$:

$$\det(\mathrm{d}\, \overline{f}|_A|_x) > 0\,.$$

Proof. Both $\overline{f}^*$ and g^* are isomorphisms

$$\overline{f}^*, g^* : K^k[D^k\big/\partial D^k] \to K^k[B^k\big/\partial B^k]$$

Let β_D be a generator of $K^k[D^k/\partial D^k]$ (obviously this is isomorphic to $\mathbb{Z}$). Then there are two cases: Either $g^*(\beta_D) = \overline{f}^*(\beta_D)$, or $g^*(\beta_D) = -\overline{f}^*(\beta_D)$. Notice how $g|_{\text{int}(D)} = \text{id}$ and thereby smooth. Hence the composition $\overline{f}^{-1} \circ g\Big|_{\text{int}(D)}$ is smooth. Assume that the Homotopie is smooth on $\text{int}(D)$ (This assumptian can by justified by Lemma 2.6.3 in [Muk15] which tells us that homotopies between maps that are smooth in certain subdomains can be made smooth over said domains.) .By the smoothnes of the homotopy the sign of the determinand of its jacobien is the same for all t and all $x \in \text{int}(D)$ and positive if we are in the first case and else negative.

$\square$

4 Morse-Theoretic Atiyah-Hirzebruch Spectralsequence

4.1 Conley index

This is just a short summary. Prooves are omitted and the goal is to remaind ourself of the definitinos.

Definition 4.1.1 (Compact invariant isolated subset)**.** For a flow $\varphi_t : M \to M$ on a locally compact metric space we call a subset $S \subseteq M$ **invariant subspace** if and only if $\varphi_t(S) = S$ for all $t \in \mathbb{R}$. For any subset $N \subseteq M$ we define the **maximal invariant subset**

$$
\begin{aligned}
I(N) &= \{x \in N | \ \varphi_t(x) \in N \forall t \in \mathbb{R}\} \\
&= \bigcap_{t \in \mathbb{R}} \varphi_t(N).
\end{aligned}
$$

A **compact invariant subset** S is called **isolated**, if a compact neighborhood N exists, such that $I(N) = S$.

Definition 4.1.2 (Index pairs)**.** Let S be an isolated compact invariant subset. A topological pair (N, L) of compact subsets of M where $L \subseteq N$ is called an **index pair of** S, if it satisfies the following:

1. $S = I(\overline{N \setminus L}) \subseteq (N \overset{\circ}{\setminus} L)$.
2. $x \in L$ and $\varphi_{[0,t]}(x) \subseteq N$ implies that $\varphi_{[0,t]}(x) \subseteq L$. We call L **positively invariant in** N. Here $\varphi_{[0,t]}(x) \coloneqq \{\varphi_{\tilde{t}}(x) | \tilde{t} \in [0, t]\}$.
3. For all $x \in N$ such that a t exists, with $\varphi_t(x) \notin N$, there exists a t' with $\varphi_{[0,t']}(x) \subseteq N$ and $\varphi_{t'}(x) \in L$.

This definition captures how the flow lines leave the invariant space S. Due to the third property we call L the **exit set**.

Definition 4.1.3 (Regular index pairs)**.** We call an index pair (N, L) **regular** if and only if the inclusion $I : L \hookrightarrow N$ is a cofibration.

Theorem 4.1.4 (Homotopy equivalence of index pairs)**.** *If S is an isolated compact invariant set and (N, L) and $(\tilde{N}, \tilde{L})$ are two index pairs of S. Then N/L and $\tilde{N}/\tilde{L}$ are homotopy equivalent as pointed spaces. This homotopie is canonical in transitive up to homotopy.*

Definition 4.1.5 (Connected Simple System)**.** A **connected simple system** consists of a cllection I_0 of pointed spaces along with a collection I_m of homotopy classes of maps between these spaces such that

(i) $\hom(X,Y) := \{[f] \in [X:Y] : [f] \in I_m\}$ consists of exactly one element for each pair of spaces $(x,y) \in I_0$, where $[X:Y]$ denotes the set of homotopy classes of maps between X and Y.

(ii) If $X,Y,Z \in I_0, [f] \in \hom(X,Y)$, and $g \in \hom(Y,Z)$, then $[g \circ f] \in \hom(X,Z)$.

(iii) $\hom(X,Y) = \{[\mathrm{id}_X]\}$ for all $X \in I_0$.

Remark 4.1.6. This definitions comes from Salamon in [Sal85].

Remark 4.1.7. Let S be a isolated, compact, invariant subset of M with respect to the Morse flow φ. Then the collections

$$I_0 := \{N/L : (N,L) \text{ is index pair for } S\}$$
$$I_m := \{[f] : f \text{ is the homotopy equivalence of index pairs from the proof of theorem 4.1.4}\},$$

make up a connected simple system. We call this $\mathcal{IP}_S$.

Definition 4.1.8 (Morphisms of Connected Simple Systems)**.** A **morphism** $\Phi : I \to Y$ between two connected simple systems $I = (I_0, I_m)$ and $J = ()J_0, J_m)$ is a collection of homotopy classes of maps between spaces io I_0 and J_0 such that

(i) For every $X \in I_0$ and $Y \in J_0$ the set $\Phi(X,Y) := \{[\varphi] \in [X:Y] : [\varphi] \in \Phi\}$ consits of exactly one element.

(ii) If $X, \bar{X} \in I_0$ and $Y, \bar{Y} \in J_0$ and if $[\varphi] \in \Phi[X,Y]$,$[f] \in \hom[\bar{X},X]$,$g \in \hom(Y,\bar{Y})$, then $[g \circ \varphi \circ f] \in \Phi(\bar{X},\bar{Y}$

Remark 4.1.9. By the second property of a morphism between connected somple systems I and J we can induce such a single map $\varphi : X \to Y$ where $X \in I_0$ and $Y \in Y_0$.

Remark 4.1.10 (Critical points as isolated compact invariant subsets)**.** Let $f : M \to \mathbb{R}$ be a Morse function on a smooth Riemannian manifold (M,g). Let $q \in \mathrm{Crit}_k(f)$. Around q we have Morse charts $\varphi : U \to T_qM$ (after identifying q_i and ∂q_i) where $\varphi(W(q \to) \cap U) \subseteq T_q^uM$ and $\varphi(W(\to q) \cap U) \subseteq T_q^sM$. With this we can define the balls:

$$\begin{aligned} D_\varepsilon^s &= \{v \in T_q^sM | \, \|v\| \le \varepsilon\}, \\ D_\varepsilon^u &= \{v \in T_q^uM | \, \|v\| \le \varepsilon\}. \end{aligned}$$

They give rise to the index pair $N_q := \varphi^{-1}(D^s \times D^u)$ and $L_q = \varphi^{-1}(D^s \times \partial D^u)$. This index pair is in fact regular, since $(N_q, L_q) \cong (D^s \times D^u, D^s \times \partial D^u) \cong (D_\varepsilon^u, \partial D_\varepsilon^u)$. Now we know the K-group of such a pair:

$$K^i(N_q, L_q) = K^i(D^k, \partial D^k) = \begin{cases} \mathbb{Z} & \text{if } i - k \in 2\mathbb{N}, \\ 0 & \text{else.} \end{cases} \tag{8}$$

Notice that for $k = 0$ we need to consider $\partial D^k = \varnothing$, to get an index pair. That's why we let ∂ denote the manifold boundary insted of the topological boundary. However, now we can identify for all k:

$$C^k(M, \mathfrak{A}, \tilde{K}^l(pt) = \bigoplus_{\mathrm{Crit}_k(f)} \mathbb{Z} \cong \bigoplus_{q \in \mathrm{Crit}_k(f)} K^k(N_q, L_q; \mathbb{Z}). \tag{9}$$

Or using the language of connected simple systems, we have the identification

define what K(IP) should mean

$$C^k(M, \mathfrak{A}, \tilde{K}^l(pt) = \bigoplus_{\mathrm{Crit}_k(f)} \bigoplus_{q \in \mathrm{Crit}_k(f)} K^k(\mathcal{IP}_q)). \tag{10}$$

.

Remark 4.1.11. Furthermore, this isomorphism

$$C^k(M, \mathfrak{A}, \tilde{K}^l(pt) = \bigoplus_{\mathrm{Crit}_k(f)} \mathbb{Z} \cong \bigoplus_{q \in \mathrm{Crit}_k(f)} K^k(N_q, L_q; \mathbb{Z}). \tag{11}$$

can be made canonically, using orientations: Let q be a critical point in index λ_q. Let $\mathfrak{b}$ be a ordered basis of $T^u_q M$. Let ψ be a refined Morse chart arround q, such that the induced basis in $T_q M = T^u_q M \oplus T^u_q M$ projects onto a a basis of the first factor $T^u_q M$ that is compatible with $\mathfrak{b}$. Using this Morse chart do define the generator of the regular index pair of q gives a unique generator.

Proof. Let ψ_1 and ψ_2 be two such Morse charts. Notice that they only differ by order of the directions and their signs, meaning that for all i there is a j such that $\psi_1^{-1}(e_i) = \pm\psi_2 - 1(e_j)$.Hence, viewed as maps into the tangent space we have that $\psi_1(N_p, L_q) = \psi_1(N_p, L_q) \subseteq T_q M$. We define $B^{\lambda_q}_\varepsilon(0) \subseteq \mathbb{R}^{\lambda_q}$ to be the sphere. The generator of $\psi_1(N_p, L_q)$ comes as the pullback of the canonical bott generator of $K^i(B^{\lambda_q}_\varepsilon(0), \partial B^{\lambda_q}_\varepsilon(0))$ via the basis map after the collapse map $c : (D^s \times D^u, D^s \times \partial D^u) \cong (D^u_\varepsilon, \partial D^u_\varepsilon)$. Let α be a generator of $\psi_1(N_p, L_q)$ induced from the basis $\mathfrak{b}_1$ that comes from ψ_1. I.e.

$$\alpha = (B_{\mathfrak{b}_1} \circ c)^* (\beta)$$

By assumption, the map $B_{\mathfrak{b}_1} \circ B^{-1}_{\mathfrak{b}}$ has determinant one. Hence it is homotopic to the identity, and this homotopy can be reduced to a well defined homotopy on the quotient space $(D^u_\varepsilon / \partial D^u_\varepsilon)$. Hence

$$\alpha = (B_{\mathfrak{b}} \circ c)^* (\beta)$$

And thereby, the generacd tor only depends on $\mathfrak{b}$ concluding the proof. □

Example 4.1.12 (A Map of K-Groups)**.** Let $f : M \to \mathbb{R}$ be a Morse Smale function on a smooth compact Riemannian manifold (M, g). Let $q \in \mathrm{Crit}_k(f)$ and $p \in \mathrm{Crit}_{k-1}(f)$. Assume for a moment we already have an isolated regular compact index pair (N_2, N_0) of $S = W(p \to q) \cup \{p, q\}$. For a $c \in (f(p), f(q))$ we set $N_1 = N_0 \cup (N_2 \cap M^c)$, where M^c is the sublevel set. Clearly now (N_2, N_1) is an index pair for q and (N_1, N_0) is a pair for p. Assuming we have regular index pairs $(N_q.L_q)$ and (N_p, L_p) for p, q we can now define a map:

$$\Delta^k(q \to p) : K^{k-1}(N_p, L_p) \to K^k(N_q, L_q)$$

as the composition:

$$K^{k-1}(N_p, L_p; \mathbb{Z}) \xrightarrow{\cong} K^{k-1}(N_1, N_0; \mathbb{Z}) \xrightarrow{\delta_{triple}} K^k(N_2, N_1; \mathbb{Z}) \xrightarrow{\cong} K^k(N_q, L_q; \mathbb{Z})$$

where δ_{triple} is the connecting homomorphism, and the other two maps are induced by the homotopic equivalence of two index pairs corresponding to the same isolated compact invariant subset. Putting all together we can define a morphism:

$$\Delta^k : \bigoplus_{p \in \mathrm{Crit}_{k-1}(f)} K^{k-1}(N_p, L_p) \to \bigoplus_{q \in \mathrm{Crit}_k(f)} K^k(N_q, L_q) \tag{12}$$

Definition 4.1.13 (Regular Index Pairs and a Filtration on M)**.** As always let $f : M \to \mathbb{R}$ be a Morse-Smale function on a compact smooth Riemannian manifold (M, g) of dimension m. Let $\varphi_t : M \to M$ be the flow given by $-\mathrm{grad} f$. For $0 \le j \le k \le m$ define:

$$W(k, j) = \bigcup_{j \le \lambda_p \le \lambda_q \le k} W(q \to p).$$

There exists a filtration

$$\varnothing =: N_{-1} \subset N_0 \subset N_1 \subset \cdots \subset N_{m-1} \subset N_m = M, \tag{13}$$

such that (N_k, N_{j-1}) is a regular index pair for $W(k, j)$ for all $0 \le j \le k \le m$. From this we can induce a filtration of conected simple systems:

$$\varnothing \to \mathcal{IP}_0 \to \mathcal{IP}_1 \to \cdots \to \mathcal{IP}_{m-1} \to \mathcal{IP}_m$$

Where $\mathcal{IP}_i$ denotes the simple system $\mathcal{IP}_{\mathrm{Crit}_i(f)}$ and the map $\mathcal{IP}_i \to \mathcal{IP}_{i+1}$ is induced from the inclusion $i : (N_i, N_{i-1}) \to (N_{i+1}, N_i)$.

we get even More! we get a list of such sequiences! maby a spectral sequence?

Proof. The proof is done by Dietmar Salamon and can be checked in [Sal85] corollary 4.4. □

4.2 The Spectral sequence

Definition 4.2.1. Let M be a smooth manifold with Morse data and

$$\varnothing =: N_{-1} \subset N_0 \subset N_1 \subset \cdots \subset N_{m-1} \subset N_m = M,$$

be a filtrations constructed in 4.1.13. Then we define the Groups

$$E_r^{p,q}(M) := \frac{\ker\left[K^{p+q}(N_{p+r-1}, N_{p-r}) \to K^{p+q}(N_{p-1}, N_{p-r})\right]}{\ker\left[K^{p+q}(N_{p+r-1}, N_{p-r}) \to K^{p+q}(N_p, N_{p-r})\right]}.$$

To define a boundary operator, we proceed as follows: The following diagram commutes (by naturality of the boundary in the triple sequence).

$$\begin{tikzcd}
K^{p+q}(N_{p+r-1}, N_{p-r}) \arrow[r, "\alpha"] \arrow[d, "\delta^*_{triple}"] & K^{p+q}(N_p, N_{p-r}) \arrow[d, "\delta^*_{triple}"] & \\
K^{p+q+1}(N_{p+2r-1}, N_{p+r-1}) \arrow[r, "\beta"] & K^{p+q+1}(N_{p+2r-1}, N_p) \arrow[r] & K^{p+q+1}(N_{p+r-1}, N_p)
\end{tikzcd}$$

Hence, $\beta \circ \delta^*_{triple}$ factors over $K^{p+q}(N_{p+r-1}, N_{p-r}) / \ker \alpha$ and because the bottom row is exact (by the triple sequence) we have get a map

$$K^{p+q}(N_{p+r-1}, N_{p-r}) / \ker \alpha \to \ker\left[K_{p+q+1}(N_{p+2r-1}, N_p) \to K^{p+q+1}(N_{p+r-1}, N_p)\right]$$

Now since $K^{p+q}(N_{p+r-1}, N_{p-r}) \supseteq \ker\left[K^{p+q}(N_{p+r-1}, N_{p-r}) \to K^{p+q}(N_{p-1}, N_{p-r})\right]$ we can restrict the domain and furthermore compose with the quotient map to get a well-defined map

$$\begin{aligned} d_r^{p,q} :\ E_r^{p,q} &\to E_r^{p+r,q-r+1} \\ [x] &\mapsto [\beta \circ \delta^*_{triple}(x)] \end{aligned}$$

Corollary 4.2.2 (The first page)**.** We start by analysing the first page

$$\begin{aligned} E_1^{p,q}(M) := \frac{\ker\left[K^{p+q}(N_p, N_{p-1}) \to K^{p+q}(N_{p-1}, N_{p-1})\right]}{\ker\left[\underbrace{K^{p+q}(N_p, N_{p-1}) \to K^{p+q}(N_p, N_{p-1})}_{\mathrm{id}}\right]} &\cong K^{p+q}(N_p, N_{p-1}) \\ &= \tilde{K}^{p+q}(N_p / N_{p-1}) \end{aligned}$$

Now since (N_p, N_{p-1}) is a regular index pair for $\mathrm{Crit}_p(f)$ we have a flow induced homotopiy equivalentce:

$$N_p / N_{p-1} \simeq \Big(\bigcup_{w \in Crit_p(f)} D^u_\varepsilon(w)\Big) / \Big(\bigcup_{w \in Crit_p(f)} \partial D^u_\varepsilon(w)\Big) \cong \bigvee_{w \in \mathrm{Crit}_p(f)} S^p$$

Here, we use the definitions from above

$$D^u_\varepsilon(w) \ := \ \varphi_w^{-1}\big(\{v \in T^u_w M \mid \|v\| \le \varepsilon\}\big),$$

and the last isomorphism is induced by orientations in the unstable manifolds. We can continue our inspection:

$$\begin{aligned} E_1^{p,q} &\cong \tilde{K}^{p+q}(N_p / N_{p-1}) \\ &\cong \bigoplus_{w \in \mathrm{Crit}_p(f)} \tilde{K}^q(pt) = C^k(M, f, g, \mathfrak{o}; \tilde{K}^q(pt)) \end{aligned}$$

In sum we have the chain of horizontal isomorphisms:

$$\begin{array}{ccccccc}
E_1^{p,q} & \xrightarrow{\alpha} & \tilde{K}^{p+q}(N_p/N_{p-1}) & \xrightarrow{\beta} & \oplus_{w\in \mathrm{Crit}_p(f)} \tilde{K}^q(pt) & \longleftarrow & C^p(M,\mathfrak{A},\tilde{K}^q(pt)) \\
\downarrow d_1^{p,q} & & \downarrow & & \downarrow & & \downarrow \\
E_1^{p+1,q} & \xrightarrow{\alpha'} & \tilde{K}^{p+1+q}(N_{p+1}/N_p) & \xrightarrow{\beta'} & \oplus_{w\in \mathrm{Crit}_{p+1}(f)} \tilde{K}^q(pt) & \longleftarrow & C^{p+1}(M,\mathfrak{A},\tilde{K}^q(pt))
\end{array}$$

By naturality of the triple boundary operator, that the second vertical map is the boundary operator. Now on the next rug we can induce two maps. From the left to make the diagram commute and from the right. Do they agree? Or asked differently, is the map induced from the d_1 map the boundary operator of morse homology?

The Main statement now is that the connecting homomorphism of K-Theory and of the long exact sequence given by the special Filtration in Morse Theory are comparable:

Theorem 4.2.3. *Given the filtration $N_{-1} \subseteq N_0 \subseteq \cdots \subseteq N_{m-1} \subseteq N_m = M$ from (13). The following diagram commutes:*

$$\begin{array}{ccc}
C^k(M,\mathfrak{A},\tilde{K}^l(pt)) & \xrightarrow{\partial^p} & C^{k+1}(M,\mathfrak{A},\tilde{K}^l(pt)) \\
\downarrow & & \downarrow \\
\bigoplus_{p\in \mathrm{Crit}_k(f)} K^{k+l}(N_p,L_p) & \xrightarrow{\Delta_{k+l}} & \bigoplus_{q\in \mathrm{Crit}_{k+1}(f)} K^{k+l+1}(N_q,L_q) \\
\downarrow & & \downarrow \\
K^{k+l}(N_k,N_{k-1}) & \xrightarrow{\delta_{triple}} & K^{p+l+1}(N_{k+1},N_k)
\end{array}$$

Proof Strategy:
We will first build up a diagramm that encapusates the connecting homomorphism from K-theory, which was build as follows: Assume that (X,Y,Z) is a topological triple. Then the first map is induced from the kollaps: $Y \to Y/Z$. Afterwards we have the connecting homomorphism of the pair (X,Y):

The K-groups of the pointed spaces X/Y and $X \cup CY$, where the latter denotes a cone above Y agree. From the latter we can quotien out X, and $X \cup CY/X \cong SY$. Hence we have the triple connecting homomophism as the composition:

$$K^{-1}(Y,Z) \to K^{-1}(Y) = K(SY) \cong K(X \cup CY/X) \to K(X \cup CY) \cong K(X/Y)$$

The triple connecting homomorphism for lower K-Groups is defined exactly the same via the identification $K^{-n}(X,Y) = K(S^n \vee (X,Y))$. For higher K-Groups we apply the natural Bott isomorphism and make use of the identity. By doing this we get the tripple homomorphism

$$\begin{tikzcd}
K^n(Y,Z) \arrow[rrr, "\delta_{triple}"] \arrow[d, Rightarrow] & & & K^{n+1}(X,Y) \arrow[d, Rightarrow] \\
K^{-n}(Y,Z) \arrow[r, "i^*"] & K^{-n}(Y,p) \arrow[r, "\delta_{double}"] & K^{-n+1}(X,Y) \arrow[r, "bott"] & K^{-n-1}(X,Y)
\end{tikzcd}$$

For the proof of the upper square, we actually proof that the following commutes:

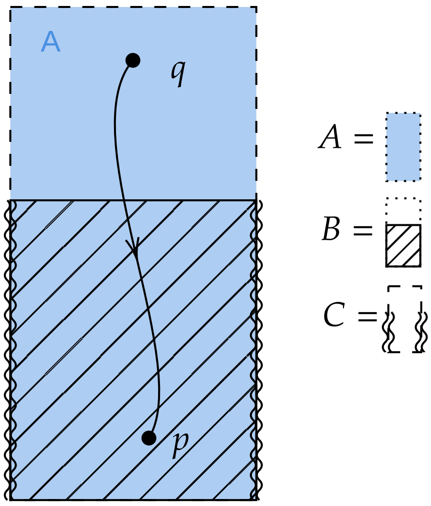
$$\begin{tikzcd}
C^k(M,\mathfrak{A},\tilde{K}^l(pt)) \arrow[r, "\partial^p"] \arrow[d] & C^{k+1}(M,\mathfrak{A},\tilde{K}^l(pt)) \arrow[d] \\
\bigoplus_{p\in\mathrm{Crit}_k(f)} K^{-k}(N_p,L_p) \arrow[r, "\Delta_k"] & \bigoplus_{q\in\mathrm{Crit}_{k+1}(f)} K^{-(k+1)}(N_q,L_q)
\end{tikzcd}$$

Diagram 14

Furthermore we assumed that $l = 0$. This doesnt change the generality since if l was even, we can apply the Bott isomorphism and else the groups are 0. Now we choose a triple (A,B,C) in M such that:

(a) (A,B) is a regular index pair for a critical point $q \in \mathrm{Crit}_{k+1}$.

(b) (B,C) is a regular index pair for a critical point $p \in \mathrm{Crit}_k$.



From the perspective of K-theory a regular index pair for a critical point is the same as a sphere with the dimension beeing the index of said critical point. To see this just use a small index pair in a Morse chart (they are all homotopic anyway), collaps the stable part and quotiend by the exit set. By doing this we get the communative diagramm:

$$\begin{tikzcd}
K^{k-1}(B,C) \arrow[r, "\delta_{triple}"] \arrow[d] & K^k(A,B) \arrow[d] \\
K^{k-1}(S^{k-1}) \arrow[r] \arrow[d] & K^k(S^{k-1}) \\
K^k(S^k) \arrow[ur] &
\end{tikzcd}$$

And as a map between K Groups of spheres, we know that this map corresponds to the question of orientation. Now we have the maps

(a) $S^k \to S \vee (B/C)$ induced from an orientation in the unstable manifold arround p together with $-\text{grad}(f)(x_i)$ where $x_i \in W(q \to p)$.

(b) $S^k \to (A/B)$ induced from an orientation in the unstable manifold arround q.

And hence the diagram

$$\begin{tikzcd} K^k(S \vee (B,C)) \arrow[rr, "\delta_{triple}"] & & K^k(A,B) \\ & K^k(S^k) \arrow[ul] \arrow[ur] & \end{tikzcd}$$

The map now factors over The K-Group of a bouquette of k-spheres generated over the connecting flow lines. In each such sphere the comparison of orientations happens.

Bevore we start proving this we need the lemma:

Lemma 4.2.4. *We work in a compact, riemanian, orientable, smooth Manifold (M,g) of dimension m together with a Morse-Smale funktion*

$$f : M \to \mathbb{R} \tag{15}$$

We assume that q is a critical point of index $k+1$ and p is a critical point of index k. we define $f(q) = b$ and $f(p) = a$ and assume that there is no critical point in $f^{-1}(a,b) \subseteq M$. We define the sub and superlevelsets

$$M^t := \{x \in M | f(x) \leq t\} \quad \text{and} \quad M_t := \{x \in M | f(x) \geq t\}, \tag{16}$$

and the constants

$$c \in (a,b) \quad , \quad \varepsilon > 0 \text{ small} \quad , \quad T > 0 \text{ big}. \tag{17}$$

With this we define the sets:

$$N_q := \{x \in M_c | f(\varphi_{-T}(x)) \leq b + \varepsilon\}, \tag{18}$$
$$L_q := \{x \in N_q | f(x) = c\}, \tag{19}$$
$$N_p := \{x \in M^c | f(\varphi_T(x)) \geq a - \varepsilon\}, \tag{20}$$
$$L_p := \{x \in N_p | f(\varphi_T(x)) = a - \varepsilon\}, \tag{21}$$

and finally:

$$C := N_p \cup N_q, \tag{22}$$
$$B := N_p \cup L_q, \tag{23}$$
$$A := L_p \cup (L_q \overset{\circ}{-} N_p). \tag{24}$$

We claim that

1. *(N_q, L_q) is a regular index pair for q .*

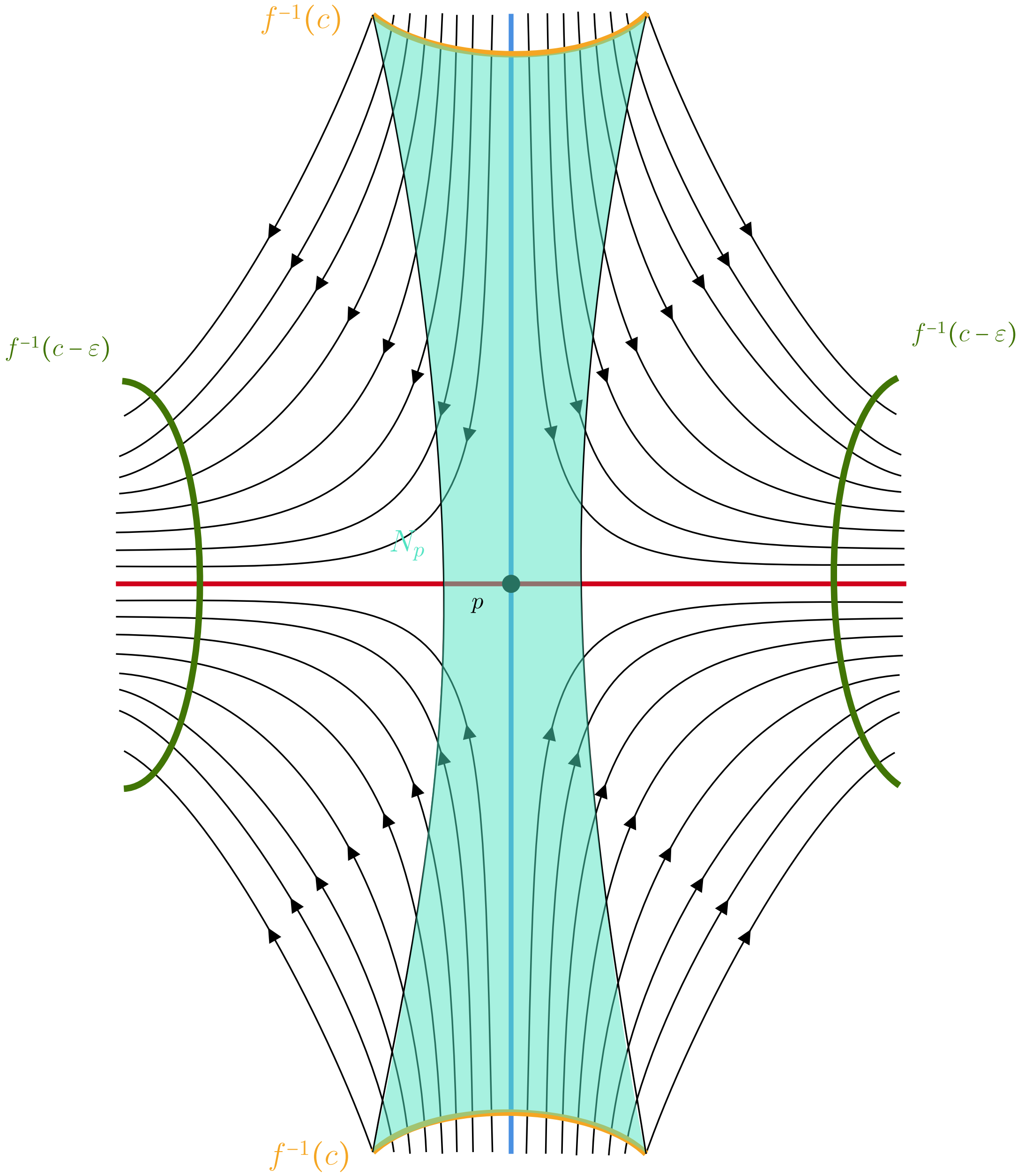
2. (C, B) *is an index pair for* q .

3. (N_p, L_p) *is a regular index pair for* p .

4. (B, A) *is an index pair for* p .

5. N_p *is a tubular neighbourhood of* $W(\to p) \cap M^c$.

Proof Strategy:
The statements 1-4 are easy to proof by the definitions. The regularity can be derived from a general fact for pairs (X, Y) in metric spaces: If Y is closed in X and there is a neighbourhood of Y that is open in X such that Y is a strong deformation retract of U. So the only thing left to show is, that N_p is a tubular neighbourhood. In [Sal90] in the proof of lemma 3.2 Salamon claims this to be true without a proof (page 119 in the attached source). Similar, in [BH04] Banyaga claims this (also without any argument). The Set N_p is sketched in the figure below. The Idea of the proove is as follows: By increasing T (or decreasing ε) we can assume N_p to be an arbitrary small neighbourhood of $W(q \to) \cap M^c$. Now using the flow we can push N_p close to p ($\varphi_N(N_p)$ for big N), where we use a tubular nieghbourhood embedding J that overlapps N_p. We now need to proof that we can reduce the radii of the fibers in the domain of J in a smooth way such that J actually becomes a embedding onto the N_p (that we pushed close to p). For this we will make use of the lambda lemma (in an improved version), which lets us induce a metric from the unstable tangend space of P into the fiberes of J (since they are transversal to the stable set). With this metric we can finally smoothly reduce the radii of our domain of J to end up with the statement, that $\varphi_N(N_p)$ is a tubular neighbourhood of a small part of the stable set of p and hence N_p also. The prove is given as an appendix in section **??**.



The same arguments work to show that N_q is a tubular neighbourhood of $W(q\to)\cap M_c$, by replacing f with $-f$. Furthermore, we can assume, that the V_j from below are fibers in the normal neighbourhood. This can be realized, since V_j intersect transversal and by trivialization we have a global fram of the bundel and hence can smoothly rescale the embedding along the frame to get that V_j is the image of the fibre above x_j.

Proof of Theorem 4.2.3. We first notice, that we have the map of triples given by the inclusion:

$$\left(W(q \to) \cap N_q \ , \ S(q \to) \ , \ S(q \to) \smallsetminus \bigcup_j V_j \right) \to (C, B, A)$$

and hence by naturalitiy we have the diagram, where the top square commutes:

$$\begin{tikzcd}
K^{-\lambda_p}\left[B, A\right] \arrow[rr, "\delta^1_{triple}"] \arrow[d, "i^*_{B,A}"] & & K^{-\lambda_q}\left[C, B\right] \arrow[d, "i^*_{C,B}"] \\
K^{-\lambda_p}\left[S(q \to), S(q \to) \smallsetminus \bigcup_j \mathrm{int}(V_j)\right] \arrow[rr, "\delta^2_{triple}"] \arrow[d, shift right, "i^*_j"'] & & K^{-\lambda_q}\left[W(q \to) \cap N_p, S(q \to)\right] \\
K^{-\lambda_p}\left[V_j, \partial V_j\right] \arrow[u, shift right, "c^*_j"'] \arrow[urr, "\delta^j_{triple}"'] & &
\end{tikzcd}$$

Diagram 25

Furthermore, in this diagram the tuple $\left(K^{-\lambda_p}\left[S(q \to), S(q \to) \smallsetminus \bigcup_j \mathrm{int}(V_j)\right], \left(c^*_j\right)_j\right)$ is a coproduct, and we define $\delta^j_{triple} \coloneqq \delta^2_{triple} \circ c^*_j$. Then we have that

$$\delta^2_{triple} = \sum_j \delta^j_{triple} \circ i^*_j$$

Since $i^*_{C,B}$ is an isomorphism (since $W(q \to \cap N_p$ is a deformation retract of C and the collaps collapses B to $S(q \to)$), we have that

$$\Delta(\alpha_q) = \delta^1_{triple}(\alpha_q) = i^*_{C,B} \circ \sum_j \delta^j_{triple} \circ i^*_j \circ i^*_{B,A}(\alpha_q) \tag{26}$$

The map δ^2_{triple} is defined as the composition:

$$\begin{tikzcd}
K^{-\lambda_p}\left[V_j, \partial V_j\right] \arrow[d, "c^*_j"] & & & \\
K^{-\lambda_p}\left[S(q \to), S(q \to) \smallsetminus \bigcup_j \mathrm{int}(V_j)\right] \arrow[rrr, "\delta^2_{triple}"] \arrow[d, "i^*_1"] & & & K^{-\lambda_q}\left[W(q \to) \cap N_p, S(q \to)\right] \\
K^{-\lambda_p}\left[S(q \to)\right] \arrow[r, "h^*_1"] & K^{-\lambda_p+1}\left[S_1 \vee S(q \to)\right] \arrow[r, "h^*_2"] & K^{-\lambda_p+1}\left[W(q \to) \cap N_p \cup C_1\left(S(q \to)\right), W(q \to) \cap N_p\right] \arrow[r, "i^*_2"] & K^{-\lambda_p+1}\left[W(q \to) \cap N_p \cup C_1\left(S(q \to)\right)\right] \arrow[u, "\beta \circ \mathrm{triv}_{C_1 S(q \to)}"']
\end{tikzcd}$$

Diagram 27

The downwards maps in the theorem are the induction of the canonical generators. This is done by the following assignement

Now the composition

$$h_1 * i_1 * \circ c_j * : K^{-\lambda_p}\left[V_j, \partial V_j\right] \to K^{-\lambda_p+1}\left[S_1 \vee S(q \to)\right]$$

can be simplified by the map

$$l^* : K^{-\lambda_p}[V_j, \partial V_j] = K^{-\lambda_p+1}[S_1 \vee (V_j, \partial V_j)] \to K^{-\lambda_p+1}[S_1 \vee S(q \to)]$$

where l is the collapsing map.

$$\begin{aligned} l : S_1 \vee S(q \to) &\to S_1 \vee (V_j, \partial V_j) \\ (t, x) &\mapsto (t, \overline{x}) \end{aligned}$$

So now we are in the following situation:

$$\begin{tikzcd}
K^{-\lambda_p}[V_j, \partial V_j] \arrow[rr, "\delta^j_{triple}"] \arrow[dr, "l_1^*"] & & K^{-\lambda_q}[W(q \to) \cap N_p, S(q \to)] \\
& K^{-\lambda_p+1}[S_1 \vee S(q \to)] \arrow[ur, "l_2"] \arrow[dr, "r^*"] & \\
K^{-\lambda_p-1}\left[\mathbb{S}^{\lambda_p+1}\right] \arrow[uu, "f_p^*"] \arrow[ur, "s^*"] \arrow[rr] & & K^{-\lambda_p-1}\left[\mathbb{S}^{\lambda_p+1}\right] \arrow[uu, "f_q^*"']
\end{tikzcd}$$

Diagram 28

The maps in the diagramm are given as follows:

- The map f_p induces α_p. Remember, that α_p comes from the canonical generator α'_p of the index pair $[B, A]$ and α'_p is defined as

$$\begin{aligned} (\varphi \circ \tilde{\psi}^{-1})^*(\beta) \text{ where } &\varphi \text{ is the homotopie equivalence of index pairs} \\ &\tilde{\psi} \text{ is a Morse chart adapted to the basis of } T_p^u M, \\ \text{and } &\beta \text{ is the bott generator of the sphere } [D^u_\varepsilon \times D^s_\varepsilon \big/ \partial D^u_\varepsilon \times D^s_\varepsilon . \end{aligned}$$

  Instead of using φ we can rescale $\tilde{\psi}$ to be a mapping to $[B, A]$ instead of first mapping to the prototype index pair $[D^u_\varepsilon \times D^s_\varepsilon, \partial D^u_\varepsilon \times D^s_\varepsilon]$. We call this rescaled Morse chart ψ. Then $\psi^{-1}(\beta) = (\varphi \circ \tilde{\psi}^{-1})^*(\beta)$ since the map $\tilde{\psi} \circ \varphi^{-1} \circ \psi^{-1}$ has the local and hence global degree one. (this can be checkt by differentiataion at the origin). Hence $\alpha'_p = (\psi^{-1})^*(\beta)$. The inclusion

$$i_p : (V_j, \partial V_j) \to (B, A)$$

  induces an isomorphism in the K-groups since N_p is a tubular neighbourhood of the contractible $W(\to p) \cap M^c$ and the contraction to the fiber V_j contracts L_p to ∂V_j (the contraction is well defined on the quotients, and for K-THeory, we only care for the qutoient spaces). Now $\alpha''_p = i_p^*(\alpha'_p)$. Define the two funktions

$$\begin{aligned} J : \mathcal{N}^{W(q \to)} W(\to p) \cap M^c &\to N_p \text{ the normal neighbourhood embedding,} \\ \nu : \mathcal{N}^{W(q \to)} W(\to p) \cap M^c &\to W(\to p) \times T_p^u M \text{ The trivialization of the normal bundle.} \end{aligned}$$

Define $\mathfrak{b}^p$ to be an ordered basis of $T_p^u M$ that is compatible to the orientation of said tanged space. Then assume that J is defined, such that

$$\left(B_{\mathfrak{b}^p}^{-1} \circ \nu_{x_j} \circ J^{-1}\right)^* (\beta) = \alpha_p'' .$$

If this is not the case just compose J with a basechage of degree one. Now α_p'' generates $K^{-\lambda_p}(V_j, \partial V_j)$. The latter is isomoprhic to $K^{-\lambda_p+1}(I \times V_j, \partial(I \times V_j))$ by definition. Now we denote the suspensision is the wedge produkt with a sphere. Hence for $K^{-\lambda_p+1}(I \times V_j, \partial(I \times V_j))$ we can induce the generator by the function

$$\begin{aligned} f_p : I \times V_j \big/ \partial(I \times V_j) &\to S \wedge S^{\lambda_p} \simeq S^{\lambda_p+1} \\ (t,x) &\mapsto \Omega\left(t, \Gamma \circ \left(B_{\mathfrak{b}^p}^{-1} \circ \nu_{x_j} \circ J^{-1}\right)(x)\right) . \end{aligned}$$

Γ maps the disk modulo its boundary to a sphere and Ω maps the suspension of the sphere to the sphere one dimension higher These Maps are given in definitino 3.0.7 and 3.0.8. We abuse the notation a bit and identify the map $\left(B_{\mathfrak{b}^p}^{-1} \circ \nu_{x_j} \circ J^{-1}\right)$ with the one defined by quotiening out the boundary. We will leave out the quotiening in the notatino from now on to make the prove easyer to read. The invested reader should check, if the quotiend maps are always well defined! Now finally we have a little definition issue: In the diagramm we have that

$$f_p^* : K^{\lambda_p-1}[\mathbb{S}^{\lambda_p+1}] \to K^{-\lambda_p}[V_j, \partial V_j] ,$$

whilst at the moment, f_p maps to $K^{-\lambda_p-1}(I \times V_j, \partial(I \times V_j))$. Hence we need to compose this f_p^* with the isomorphism that comes from the suspension axiom:

$$K^{-\lambda_p}(X) \cong K^{-\lambda_p+1}(SX) .$$

And then add the bott isomorphism to rais th eindex by two. By abuse of notation we dont write that idetification and rather think of the top left corner as $K^{-\lambda_p+1}(I \times V_j, \partial(I \times V_j))$.

- Again, we want f_q to be the map that induces the generator of $K^{-\lambda_q}[W(q \to) \cap N_q, S(q \to)]$ which comes from the canonical generator of $[C, B]$ together with the inclusion. Again instead of first introducing the generator on the prototype index pair, we can use E_q^{u-1} instead of the Morse chart. Again we can argue here, that the local degree of changing between the two is one, since E_q^{u-1} is a refined morse chart which can be choosen to be adapted to the ordered basis $\mathfrak{b}^q$ of $T_q^u M$.´ Hence we can define f_q to be the quotient map:

$$\begin{aligned} W(q \to) \cap N_q \big/ S(q \to) &\to S^{\lambda_p+1} \\ x &\mapsto \Gamma \circ B_{\mathfrak{b}^q}^{-1} \circ E_q^{u-1}(x) . \end{aligned}$$

Here Γ maps the disk modulo its boundary to the sphere. Hence $f_q = \left(B_{\mathfrak{b}^q}^{-1} \circ E_q^{u-1}\right)^*$: K

- The map l_1 is the collapsing map

$$S_1 \wedge S(q \to) \to S_1 \wedge (V_j / \partial V_j)$$

- The map l_2 is not topologically induced.
- The map s is given by $f_p \circ l_1$. Again we abuse the notation and leave out the suspension isomorphism when

$$s^* : K^{-\lambda_p+1}[S_1 \wedge S(q \to)] \to K^{-\lambda_p-1}[\mathbb{S}^{\lambda_p+1}] .$$

- To find the map r that fits the diagram (to make it commute) is the next goal. First we need to dig into l_2: From diagram 27 we know that $l_2 = \Sigma \circ \mathrm{triv}_{C_1 S(q\to)} \circ i_2^* \circ h_2^*$, Where Σ denotes the suspension isomorphism. Hence, if the function r^* makes the diagram commutative we need to ensure, that $f_q^* \circ r^* = l_2$. If we define the function

$$col : W(q \to) \cap N_p \cup C_1 S(q \to) \to S_1 \wedge S(q \to)$$

to be the collision, of $W(q \to) \cap N_p$ to a point, then $i_2^* h_2^* = col^*$. Now while the function $\mathrm{triv}_{C_1 S(q\to)}$ is not topologically induced, its inverse is. Lets call the collaps of $C_1 S(q \to)$:

$$k_1 : (W(q \to) \cap N_p) \cup C_1 S(q \to) \to (W(q \to) \cap N_p / S(q \to))$$

π_1. Then since Σ is a natural transformation we have

$$f_q^* \circ r^* = l_2 \Leftrightarrow k_1^* \circ f_q^* \circ r^* = \Sigma \circ col^*$$

Now all maps are topologically induced except Σ and the latter is just used to geht the index in the K group right. Hence, if r would be such that

$$r \circ f_q \circ c_1 = col ,$$

then the requirement on the induced maps would be satisfied. We first however need to formulate a little lemma:

Lemma 4.2.5. *Let $S^k := \{x \in \mathbb{R}^{k+1} | \, \|x\| = 1\}$, $H_+ := \{x \in S^k | x_0 \geq 0\}$ denote the upper hemissphere, H_- the lower hemissphere, $\pi : S^k \to D^k$ with $\pi(x_0, \ldots, x_k) = (x_1, \ldots x_k)$ be the projection and Λ be the homeopmorphism from $D^k / \partial D^k \to S^k$. Let furthermore $c_{Eq} : S^k \to \overline{H_+} \vee \overline{H_-}$ be the collaps of the equator, where $\overline{H_+}$ denotes the upper hemisphere modulo the equator and $\overline{H_-}$ is defined analog. Let finally $c_2 : \overline{H_+} \vee \overline{H_-} \to \overline{H_+}$ be the collaps of the second summand and analogusly $c_1 : \overline{H_+} \vee \overline{H_-} \to \overline{H_-}$. Then we have that $\Lambda \circ \pi \circ c_2 \circ c_{Eq} \simeq -\Lambda \circ \pi \circ c_1 \circ c_{Eq}$, where de minus denotes any map homeomorphism of negative degree.*

Proof. This lemma is the concrete version of a standart lemma that is often formulated along the lines: Let $q : S^k \to S^k \vee S^k$ be the collaps of the equator and p_i be the projection to the i-th summand, then $p_1 \circ q = -p_2 \circ q$. □

Now we need to transport our situation to fit the lemma. For this we will use the maps defined in the lemma above, and need a bunch of maps again:

- Firts, the map ρ is given by

$$\begin{aligned}\rho : W(q \to) \smallsetminus \{q\} &\to W(q \to) \cap f^{-1}(c) \smallsetminus W(q \to) \smallsetminus \{q\}\\ x &\mapsto \varphi_{\tau(x)}(x)\,,\end{aligned}$$

  and

$$\begin{aligned}\tau : W(q \to) \smallsetminus \{q\} &\to \mathbb{R}\\ x &\mapsto \tau(x) \text{ such that } f \circ \varphi_{\tau(x)}(x) = c\,.\end{aligned}$$

- $\Phi : S^{\lambda_q} \to W(q \to) \cap N_p \cup C_1 S(q \to)$ is defined with a case distingtion. We define

$$\begin{aligned}\Phi|_{H_+}(x) &= E_q^u \circ B_{\mathfrak{b}^q}^{-1} \circ \pi(x)\\ \Phi|_{H_-}(x) &= (\mathrm{e}^{\tau(y)}, \rho(y)) \text{ where } y = E_q^u \circ B_{\mathfrak{b}^q}(\pi_-(x))\,.\end{aligned}$$

  Notice, how this function is continuous and does not depend on the choice taken in the definition of ρ'.
- $\Psi : S^{\lambda_q} \vee S^{\lambda_q} \to (W(q \to) \cap N_p / S(q \to)) \vee \mathbb{S} \wedge S(q \to)$ is given by mapping $x \mapsto \overline{\Phi(x)}$. The well definition is easy to see.
- p_i for $i = 1, 2$ denotes the projection onto the first or second summand in a pointed sum.
- $C_{S(q\to)} : W(q \to) \cap N_q \cup C_1 S(q \to) \to (W(q \to) \cap N_p / S(q \to)) \vee \mathbb{S} \wedge S(q \to)$ is the collaps of $S(q \to)$.
- $R : \mathbb{S}^{\lambda_q} \to \mathbb{S} \wedge S(q \to)$ is defined to be $col \circ \Phi$

This gives the commutative diagram below, where by the lemma above the upper path is the negative of the lower path.

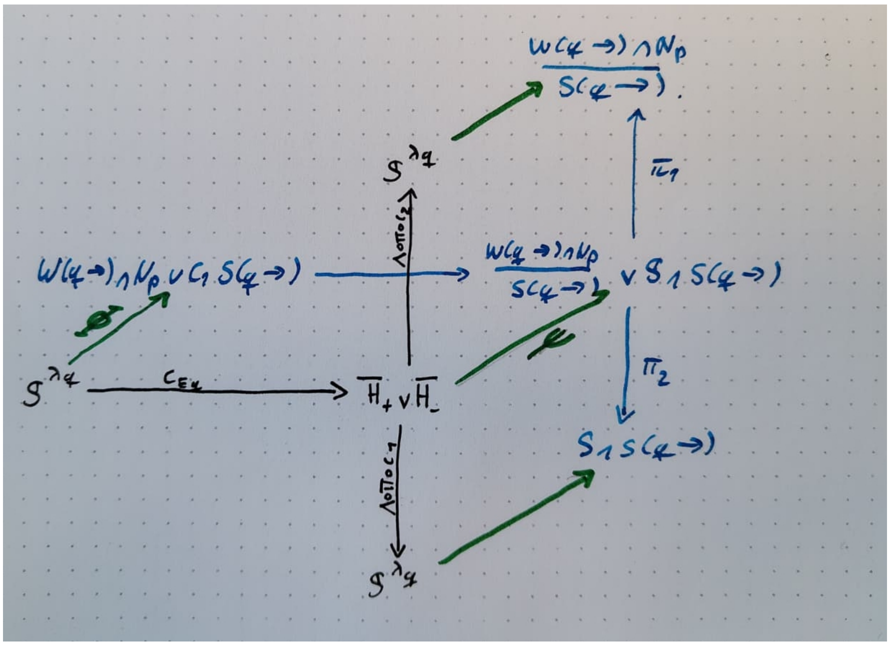


Diagram 29

Now define $r := -R$. For example apply R after the reflection $(x_0, \dots x_{\lambda_q}) \mapsto (-x_0, \dots, x_{\lambda_q})$. Then by the lemma above and commutativity we have up to homotopy:

$$\begin{aligned} r \circ f_q \circ k_1 &= -R \circ f_q \circ k_1 \circ \Phi \circ \Phi^{-1} \\ &= -R \circ \Lambda \circ \pi \circ c_2 \circ C_{Eq} \circ \Phi^{-1} \\ &= R \circ \underbrace{\Lambda \circ \pi \circ c_1 \circ C_{Eq}}_{\simeq \mathrm{id}} \circ \Phi^{-1} \\ &= col\,. \end{aligned}$$

With r we finally have all the maps. Now we want to find out what degree the composition $s \circ r$ has. For this we calculate the local degree in some point by checking the sign of the determinant of the Jacobean. This strategy works out since we are in the situation of lemma 3.0.9. Bevor we start differentiating, we expand the map arround $x \in \mathbb{R}^{\lambda_q}$, such that $x' := E_q^u \circ B_{\mathfrak{b}^q}(x) \in \rho{-}1(x_j)$, meaning that x' is on the flow line towards x_j and assume that $\tau(x') = 1$.

$$
\begin{aligned}
&\pi_+ \circ s \circ r \circ \pi_+^{-1}(x_1, \dots, x_{\lambda_q}) \\
&= \pi_+ \circ s \circ R(-x_0, x_1, \dots, x_{\lambda_q}) \quad x_0 > 0 \\
&= \pi_+ \circ s \circ col \circ \Phi|_{H_-}(-x_0, x_1, \dots, x_{\lambda_q}) \\
&= \pi_+ \circ f_p \circ l_1 \circ col\left(e^{\tau(x')}, \rho(x')\right) \qquad col \text{ and } l_q \text{ are the identity arround } \left(e^{\tau(x')}, \rho(x')\right) \\
&= \pi_+ \circ \Omega\Big(e^{\tau(E_q^u \circ B_{\mathfrak{b}^q}(x))}, \Gamma \circ \underbrace{B_{\mathfrak{b}^p}^{-1} \circ \nu_{x_j} \circ J^{-1} \circ \rho \circ E_q^u \circ B_{\mathfrak{b}^q}(x)}_{:=Z(x)}\Big) \\
&= \sin(\pi e^{\tau(E_q^u \circ B_{\mathfrak{b}^q}(x))}) \cdot \begin{pmatrix} -\cos(\pi \cdot \|Z(x)\|) \\ \sin(\pi \cdot \|Z(x)\|) \cdot \frac{Z(x)}{\|Z(x)\|}) \end{pmatrix}
\end{aligned}
$$

Here Z is a funktion $Z : \mathrm{int}(I \times D^{\lambda_q}) \to \mathbb{R}^{\lambda_q - 1}$ and we can calculate its jacobean Notice how $Z(y) = 0$ where y is the same as above. We only care for the sign of its jacobean, so we will manipulate the funktion in a way that might change the function, but not the sign of ots Jacobean determinant. Let $Q_y : V \to T_y V$ be the canonical map, that identifies a linear space with the tangendspace of a point from inside said space. Now we calculate:

$$
\begin{aligned}
&Q_0^{-1} \circ \mathrm{d}\left(B_{\mathfrak{b}^p}^{-1} \circ \nu_{x_j} \circ J^{-1} \circ \rho \circ E_q^u \circ B_{\mathfrak{b}^q}\right)\big|_y \circ Q_y(e_i) \\
&= Q_0^{-1} \circ \mathrm{d}\left(B_{\mathfrak{b}^p}^{-1} \circ \nu_{x_j} \circ J^{-1} \circ \rho \circ E_q^u\right)\big|_{B_{\mathfrak{b}^q}(y)} \circ Q_{B_{\mathfrak{b}^q}(y)} \circ B_{\mathfrak{b}^q}(e_i) \\
&= Q_0^{-1} \circ \mathrm{d}\left(B_{\mathfrak{b}^p}^{-1} \circ \nu_{x_j} \circ J^{-1}\right)\big|_{x_j} \circ \mathrm{d}\left(\rho \circ E_q^u\right)\big|_{B_{\mathfrak{b}^q}(y)} \circ Q_{B_{\mathfrak{b}^q}(y)} \circ B_{\mathfrak{b}^q}(e_i)
\end{aligned}
$$

Now the term $\mathrm{d}\left(\rho \circ E_q^u\right)\big|_{B_{\mathfrak{b}^q}(y)} \circ Q_{B_{\mathfrak{b}^q}(y)} \circ B_{\mathfrak{b}^q}$ maps e_1 to zero. and $(e_2, \dots e_{\lambda_q})$ induces a basis in $T_{x_j} V_j$ that is compatible with a basis $(\mathfrak{d}_i)_{i=2,\dots\lambda_q}$ that satisfies the following: Adding $-\mathrm{grad}_{x_j}(f)$ as the first basis vector to $(\mathfrak{d}_i)_{i=2,\dots\lambda_q}$ gives a basis of $T_{x_j} W(q \to)$ that is compatible with the basis induced from $\mathfrak{b}^q$ that appears in the counting of flow lines in the Morse boundary operator. This was proven earlyer in lemma 2.0.24. Now we can look at the first part:

$$
\begin{aligned}
Q_0^{-1} \circ \mathrm{d}\left(B_{\mathfrak{b}^p}^{-1} \circ \nu_{x_j} \circ J^{-1}\right)\big|_{x_j} &= \left(\mathrm{d}J|_{x_j} \circ \mathrm{d}\nu_{x_j}^{-1}\big|_0 \circ \mathrm{d}B_{\mathfrak{b}^p}|_0 \circ Q_0\right)^{-1} \\
&= \Big(\underbrace{\mathrm{d}J|_{x_j} \circ Q_0}_{=\mathrm{id}_{T_{x_j} V_j}} \circ \nu_{x_j}^{-1} \circ B_{\mathfrak{b}^p}\Big)^{-1} \\
&= \left(\nu_{x_j}^{-1} \circ B_{\mathfrak{b}^p}\right)^{-1}
\end{aligned}
$$

Notice how $\nu_{x_j}^{-1} \circ B_{\mathfrak{b}^p}$ induces a basis on $T_{x_j} V_j$ from a basis in $T_p^u M$. Comparing this basis to the basis $(\mathfrak{d}_i)_{i=2,\dots\lambda_q}$ is what assigned the sign to the flow line in the Morse boundary operator. By the calculations we know that the Jacobean of the funktion Z

$$\mathrm{d}Q_0^{-1} \circ \mathrm{d}\left(B_{\mathfrak{b}^p}^{-1} \circ \nu_{x_j} \circ J^{-1} \circ \rho \circ E_q^u \circ B_{\mathfrak{b}^q}\right)\Big|_y \circ Q_y = \begin{pmatrix} 0 & \tilde{Z} \end{pmatrix},$$

where $\tilde{Z}$ is a λ_{q-1} matrix with the sign of its determinant being given by the sign in the Morse boundary operator n_γ.

Now we can calculate the differential

$$\begin{aligned} &\mathrm{d}\Big(\pi_+ \circ s \circ r \circ \pi_+^{-1}(x_1, \ldots, x_{\lambda_q})\Big)\Big|_x \\ &= \mathrm{d}\left(\sin(\pi e^{\tau(E_q^u \circ B_{\mathfrak{b}^q}(x))}) \cdot \begin{pmatrix} -\cos\left(\pi \cdot \|Z(x)\|\right) \\ \sin\left(\pi \cdot \|Z(x)\|\right) \cdot \frac{Z(x)}{\|Z(x)\|}) \end{pmatrix}\right)\Bigg|_x \end{aligned}$$

To keep it readable we will first calculate

$$\mathrm{d}\left(\sin(\pi e^{\tau(E_q^u \circ B_{\mathfrak{b}^q}(x))})\right)\Big|_x = \cos(\pi \cdot \mathrm{e}) \cdot \pi \mathrm{e} \cdot \mathrm{d}\tau|_{E_q^u \circ B_{\mathfrak{b}^q}(x)} = \Big(\underbrace{\cos(\pi \cdot \mathrm{e}) \cdot \pi \mathrm{e}}_{<0}, 0, \ldots, 0\Big)$$

In this calculation we used that x was choosen such that $\tau \circ E_q^u \circ B_{\mathfrak{b}^q} = 1$ and $\mathfrak{b}^q$ was such that $\mathrm{d}E_q^u\big|_y \circ Q_y(\mathfrak{b}_i^q)$ equals $-\mathrm{grad}_y(f)$ for $i = 1$ and for $i > 1$ it equals $\mathrm{d}\varphi_{\tau(y)}\big|_y^{-1}(w_i)$ where (w_i) denotes a basis of $T_{x_j}V_j$. Hence applying the product rule to the differential above yealds a sum with the first summond being

$$(-1, 0, \ldots, 0) \cdot \Big(\underbrace{\cos(\pi \cdot \mathrm{e}) \cdot \pi \mathrm{e}}_{<0}, 0, \ldots, 0\Big)^\top = \begin{pmatrix} a & 0 \\ 0 & 0 \end{pmatrix} \quad \text{where } a > 0\,.$$

Here we just used, that $Z(x) = 0$. The second summand is:

$$\underbrace{\sin(\pi \cdot e)}_{>0} \begin{pmatrix} \sin(0) \cdot * = 0 \\ \pi \cdot 1_{\lambda_q \times \lambda_q} \circ \mathrm{d}Z|_x \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & b \cdot Z \end{pmatrix} \quad \text{where } b > 0)\,.$$

Here we used that

$$\mathrm{d}\frac{\sin(\pi\,\|x\|)}{\|x\|} \cdot x \Bigg|_{x \to 0} = \pi \cdot 1_{\lambda_q \times \lambda_q}$$

But with this we have the jacobean

$$\mathrm{d}\Big(\pi_+ \circ s \circ r \circ \pi_+^{-1}(x_1, \ldots, x_{\lambda_q})\Big)\Big|_x = \begin{pmatrix} a & 0 \\ 0 & bZ \end{pmatrix},$$

and know that its determinant has a positive that equals n_γ which is the assigned sign to the flow line containing x_j in the Morse Boundary operator. Or said differently: The local and hence global degree of the map $r^* \circ s^*$ equals n_γ. With this we can calculate

$$\delta^j_{triple}(\alpha_p) = \sum_j \delta^j_{triple} \circ f_p^*(\beta) = n_\gamma f_q^*(\beta) = n_\gamma(\alpha_q)$$

Furthermore: $\Delta(\alpha_p) = \delta^2_{triple}(\alpha_p) = \sum_{\gamma \in ,} n_\gamma(\alpha_q)$. This concludes the prove. □

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